# Exogenous Features in Intraday Forecasts of Closing-Half-Hour Variance

Bullet-point draft of the full paper

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Working draft, 18 October 2026 — not for citation

**Abstract**

We investigate whether exogenous predictors improve intraday forecasts of a futures-based (E-mini) proxy for the realized variance of the S&P 500 over the closing half-hour interval (15:30–16:00 ET). Specifically, we estimate each model across all half-hour intervals and evaluate the forecast issued at 15:30 for the closing half hour over the out-of-sample period from 3 January 2020 to 30 April 2024 ($T_{\text{oos}}$ = 1,082 trading days). We compare a benchmark heterogeneous autoregressive (HAR) predictor set against an augmented predictor set that incorporates exogenous series. Under the QLIKE loss, the full predictor set yields lower average out-of-sample loss than the HAR benchmark across all 6 intraday model classes, with 4 of the 6 reductions statistically significant at the 5% level. Furthermore, although every intraday specification remains less accurate under QLIKE than a HAR-type model in the spirit of Corsi (2009), estimated by OLS on the closing-interval observations of the target's trailing averages over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples plus calendar columns, incorporating the exogenous predictors narrows this gap across all model classes. In an economic application to zero-days-to-expiration (0DTE) S&P 500 straddles on expiration days, a sign-based trading strategy achieves a higher Sharpe ratio under the full predictor set than under the HAR benchmark across every intraday model class.

**Notes.** Colored notes are draft comments, not paper text.
**Red**: two live statements disagree.
**Orange**: a sentence does not match the computation behind a printed number.
**Blue**: a reader cannot recover the input or the rule.
The first line is the decision. The second line is the problem. The rest is the exposition.

**E34.** Decision: base 5 across the paper (user, 2026-10-19).
Problem: the pooled trees (LightGBM, XGBoost and the random forest, on every predictor set) and the intraday OLS on the HAR benchmark (note a of Table 1) were fitted on the base-2 panel, horizons $\{1, 2, 4, \ldots, 2048\}$. Every number built from them still comes from that design: the counts of intraday model classes in this abstract and in the introduction (6 classes, 4 significant), the pooled tree and intraday OLS rows of Table 1, and the pooled comparisons of Sections 6 and 7 and of the appendix.
The pooled ridge, lasso and elastic net printed here were already fitted at base 5 (experiments/pool_linear_pretune.py uses the production horizon set). The pooled trees are not refitted (user, 2026-10-19: not in this pass); the base-5 design for a refit exists (experiments/pooled_trees_base5.py, design stage; results/pooled_trees_base5_2026-10-19_48 and _13 hold its column lists). The intraday OLS is replaced by the pooled OLS of the linear reruns. The generators of these numbers (make_rehaul_results.py, make_rehaul_oos_table.py, make_rehaul_economics.py) still read results/pooled_trees_2026-10-07; switch them after the refits.

# 1 Introduction

- **Research question.** We investigate whether incorporating exogenous predictors improves intraday forecasts of a futures-based (E-mini) proxy for the realized variance of the S&P 500 over the closing half-hour interval.
  - Specifically, at the 15:30 ET forecast origin on trading day $t$, we seek to predict $\mathrm{RV}_t$, which denotes the realized variance over the 15:30–16:00 closing interval (see Section 3).
  - Each intraday model is estimated across all $M = 48$ half-hour intervals of the trading day, and we evaluate its out-of-sample predictions on the forecast issued at 15:30 for the closing half hour.
  - To isolate the contribution of the additional covariates, we compare the full predictor set against the HAR benchmark; the exogenous series constitute the exact difference between the two sets (see Section 4).
- **Motivation for the HAR baseline.** Following Corsi (2009), the standard HAR regression models realized volatility solely as a linear function of its own past lags. Consequently, it omits a rich set of exogenous variables available by 15:30, including higher-order realized moments, liquidity measures, the cross-section of constituent returns, news and social-media sentiment, option-implied volatility, volatility demand indicators, and scheduled macroeconomic and earnings announcements.
- **Empirical design.** For each intraday interval $(t, i)$, we consider a single forecast mapping of the form

  $$\hat{y}_{t,i} = f_{m,\mathcal{X},\mathcal{I}}(\boldsymbol{x}_{t,i}),$$

  indexed by the model class $m$ and the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$.
  - The primary empirical results reported in Section 6 focus on the pooled intraday estimation scheme, $\mathcal{I} = \{1, \ldots, M\}$.
  - As our primary benchmark forecast, we employ the ordinary least squares (OLS) regression on the HAR benchmark predictor set, estimated exclusively on the closing interval (the closing-interval HAR benchmark; see Sections 5.2 and 5.3).
  - We evaluate predictive accuracy via the average QLIKE loss over the out-of-sample evaluation span $\mathcal{T}_{\mathrm{oos}}$ from 3 January 2020 to 30 April 2024 ($T_{\mathrm{oos}} = 1{,}082$ trading days; see Section 5.3).
- **Main findings.**
  - Incorporating the full predictor set reduces the average out-of-sample QLIKE loss relative to the HAR benchmark across every intraday model class (6 classes in total, with 4 of these reductions statistically significant at the 5% level; see Section 6). The tree-based ensembles exhibit the largest relative improvements, while among the linear specifications, the elastic net achieves a statistically significant reduction.
  - Every pooled intraday forecast incurs a higher average QLIKE loss than the closing-interval HAR benchmark. While this performance gap is statistically insignificant at the 5% level for the linear models, it is significant for the tree ensembles. Notably, augmenting the intraday models with the full predictor set narrows this gap across all model classes (see Section 6.4).
- **Economic application.** To assess the economic value of the forecasts on trading days with a same-day SPXW expiration, we construct a trading strategy that takes a long or short position in a zero-days-to-expiration (0DTE) at-the-money straddle based on the sign of the difference between the variance forecast and option-implied variance (see Section 7). Using

the full predictor set increases the annualized Sharpe ratio of this strategy relative to the HAR benchmark in every intraday model class (see Section 7.3).

**N15c.** Problem: "every intraday model class" no longer holds under the per-year Sharpe ratio (pooled random forest $\Delta\text{SR} = -0.19$); see note N15c in Section 7.

- **Paper organization.** The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and target construction, and Section 4 details the benchmark and exogenous predictors. Section 5 presents the forecasting models and rolling estimation protocol. Section 6 reports the out-of-sample forecasting results. Section 7 evaluates the economic value of the forecasts in the 0DTE straddle strategy, and Section 8 concludes.

## 2 Related work

- **Realized variance and the HAR model.** Following Corsi (2009), the heterogeneous autoregressive (HAR) model specifies the conditional expectation of next-day realized variance as a linear function of past realized variances aggregated over daily, weekly, and monthly horizons: $\mathbb{E}_t[\text{RV}_{t+1}^{(d)}] = \beta_0 + \beta_d \,\text{RV}_t^{(d)} + \beta_w \,\text{RV}_t^{(w)} + \beta_m \,\text{RV}_t^{(m)}$, where $\mathbb{E}_t$ denotes the conditional expectation given information available at the forecast origin on day $t$, and $\text{RV}_t^{(d)}$, $\text{RV}_t^{(w)}$, and $\text{RV}_t^{(m)}$ denote the average daily realized variance over the preceding 1, 5, and 22 trading days, respectively. Because all three components represent daily realized variances aggregated at the daily, weekly, and monthly horizons, forecasting the realized variance of a specific intraday interval lies outside this standard daily specification.
  See Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), and Andersen et al. (2006).
  - In a multivariate setting, the panel HAR framework of Bollerslev et al. (2018) estimates a single pooled regression jointly across a cross-section of assets.
  - For the theoretical foundations of realized variance, see Barndorff-Nielsen and Shephard (2002).
- **Diurnal pattern.** It is well documented that intraday return volatility exhibits a pronounced U-shaped pattern across the trading session. To account for this periodicity, standard practice scales intraday realized variance by a diurnal factor estimated from the corresponding clock-time interval on prior trading days.
  See Andersen and Bollerslev (1997) on intraday periodicity, Engle and Gallo (2006) on intraday range and volatility measures as indicators of daily volatility, and Hizmeri and Bevilacqua (2025) on morning volatility uncertainty.
  - See also Taylor and Xu (1997) on diurnal volatility adjustments and Engle and Sokalska (2012) on multiplicative decompositions of intraday volatility.
- **0DTE options and the variance risk premium.** The variance risk premium, defined as the difference between option-implied variance and expected realized variance, $\mathbb{E}_t^{\mathbb{Q}}[\text{RV}_t] - \mathbb{E}_t^{\mathbb{P}}[\text{RV}_t]$, under the risk-neutral ($\mathbb{Q}$) and physical ($\mathbb{P}$) measures (where $\text{RV}_t$ denotes the realized variance through expiration), is empirically positive on average.
  See Hizmeri and Bevilacqua (2025), as well as the foundational option-pricing frameworks of Black and Scholes (1973) and Merton (1973).
  - On zero-days-to-expiration (0DTE) options, see Bandi et al. (2023), Beckmeyer et al.

(2023), Brogaard et al. (2023), Dim et al. (2024), Vilkov (2026), and Almeida et al. (2025); on the variance risk premium more broadly, see Carr and Wu (2009) and Bollerslev et al. (2009).

- **Machine learning for volatility and returns.** A growing literature models conditional return and volatility dynamics via flexible nonlinear mappings $f(\boldsymbol{x}_t)$ of a high-dimensional predictor vector $\boldsymbol{x}_t$.
  See Gu et al. (2020) for return prediction and Christensen et al. (2023) for realized volatility forecasting; for gradient-boosted decision trees, see Ke et al. (2017) (LightGBM) and Chen and Guestrin (2016) (XGBoost); for Bayesian hyperparameter optimization, see Akiba et al. (2019) (Optuna).
  - For ensemble methods, see Breiman (2001) on random forests and Friedman (2001) on gradient boosting; for machine-learning applications to realized volatility and intraday commonality, see Bucci (2020), Rahimikia and Poon (2020), and Zhang et al. (2024).
- **Shrinkage with many weak signals.** In high-dimensional linear regressions, penalized estimators regularize the coefficient vector via

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2)$$

  for a response vector $\boldsymbol{y}$, a predictor matrix $\boldsymbol{X}$, and an overall penalty strength $\lambda$, nesting ridge regression ($\alpha = 0$) and the lasso ($\alpha = 1$) as special cases; multi-penalty ridge extensions assign distinct regularization weights across feature blocks via $\lambda \sum_j \omega_j \beta_j^2$, where the penalty weight $\omega_j$ is held constant within each block.
  See Giannone et al. (2021) on dense versus sparse representations, Shen and Xiu (2025) on learning from many weak signals, Kozak et al. (2020) on shrinking the cross-section of returns, Hoerl and Kennard (1970) on ridge regression, Tibshirani (1996) on the lasso, Zou and Hastie (2005) on the elastic net, and van de Wiel et al. (2021) and van Wieringen (2023) on multi-penalty ridge estimation.
  - See also Zou (2006) on the adaptive lasso and feature-specific penalty weights, and Friedman et al. (2010) on coordinate-descent estimation under the elastic-net parametrization.

# 3 Data

- **Data sources.** We group the exogenous covariates into eight distinct categories. Specifically, the pooled intraday specifications employ the 47 series across all categories, the FOMC group included. A complete list and definition of each individual series is provided in Appendix Table 5.
  - **Realized moments** (7 series). We obtain intraday data on the front-month E-mini S&P 500 futures contract (defined as the most actively traded contract), which trades nearly 24 hours a day from 1998 through 2024, aggregated onto a 30-minute grid. Here, $t = 1, \ldots, T$ indexes trading days and $i = 1, \ldots, M$ indexes the half-hour intervals within day $t$, each ending on a half-hour timestamp; in particular, $i = c$ denotes the 15:30–16:00 ET closing interval. Each realized moment for interval $(t, i)$ is computed by summing the corresponding one-minute quantity across all non-missing minutes within that half hour.
  - Let $r_{t,i,k}$ denote the one-minute log-price change of the front-month contract for the $k$-th non-missing minute in interval $i$ of day $t$, where $k = 1, \ldots, N_{t,i}$ and $N_{t,i} \leq 30$. Following

Andersen et al. (2003), we define the realized variance over interval $(t, i)$ as

$$\mathrm{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2.$$

– We construct the 7 exogenous realized-moment series as analogous one-minute sums over interval $(t, i)$ (note that $\mathrm{RV}_{t,i}$ above serves as our forecast target and is not counted among the exogenous series):

$$\texttt{sumret}_{t,i} = \sum_k r_{t,i,k}, \quad \texttt{sumabsret}_{t,i} = \sum_k |r_{t,i,k}|, \quad \texttt{sumret3}_{t,i} = \sum_k r_{t,i,k}^3,$$
$$\texttt{sumret4}_{t,i} = \sum_k r_{t,i,k}^4, \quad \texttt{sumpret2}_{t,i} = \sum_k r_{t,i,k}^2 \, \mathbf{1}\{r_{t,i,k} > 0\},$$
$$\texttt{sumbipow}_{t,i} = \sum_{k\geq 2} |r_{t,i,k}| \, |r_{t,i,k-1}|, \quad \texttt{sumautocov}_{t,i} = \sum_{k\geq 2} r_{t,i,k} \, r_{t,i,k-1}.$$

Specifically, the third sum corresponds to the unnormalized realized skewness of Amaya et al. (2015) (that is, prior to dividing by $\mathrm{RV}_{t,i}^{3/2}/\sqrt{N_{t,i}}$), and the fourth sum is the unscaled realized quarticity (without the factor $N_{t,i}/3$). Furthermore, $\texttt{sumpret2}_{t,i}$ denotes the positive realized semivariance of Barndorff-Nielsen et al. (2010), $\texttt{sumbipow}_{t,i}$ denotes the unscaled bipower variation of Barndorff-Nielsen and Shephard (2004) (without the $\pi/2$ normalization factor), and $\texttt{sumautocov}_{t,i}$ denotes the first-order realized autocovariance.

– **Liquidity** (14 series). From the front-month futures contract, we include total contract volume traded during the half hour as well as the number of observed one-minute intervals $N_{t,i}$ (`numobs`). In addition, using intraday data on S&P 500 constituents from 1993 to 2024, we construct both equal-weighted and value-weighted cross-sectional aggregates of share turnover, buyer-initiated turnover, seller-initiated turnover, the effective spread, the quoted spread, and order-flow imbalance. Here, constituent turnover is defined as the number of shares traded during the half hour divided by prior-day shares outstanding, while buyer- and seller-initiated turnover aggregate trades according to their estimated trade direction. The effective spread measures the average percentage deviation between the transaction price and the prevailing mid-quote, and the quoted spread measures the average percentage bid–ask spread. Equal-weighted aggregates take simple cross-sectional averages across constituents, whereas value-weighted aggregates weight each constituent by its prior-day market capitalization. For each weighting scheme (`ofi_ewstock` and `ofi_vwstock`), we define order-flow imbalance as

$$\mathrm{OFI}_{t,i} = \frac{\mathrm{Buy}_{t,i} - \mathrm{Sell}_{t,i}}{\mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i}} \in [-1, 1], \qquad \text{missing if } \mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i} = 0,$$

where $\mathrm{Buy}_{t,i}$ and $\mathrm{Sell}_{t,i}$ denote the aggregate buyer- and seller-initiated turnover in interval $i$ of day $t$, respectively.

– **Equal-weighted market** (6 series) and **value-weighted market** (6 series). For each weighting scheme described above, we compute six cross-sectional constituent return moments: realized variance, realized skewness, realized quarticity, absolute variation, bipower variation, and positive semivariance. In contrast to the futures moments, which are constructed from one-minute log-price changes, these constituent moments are computed from 5-minute log returns. Moreover, the constituent moment groups omit signed

returns and realized autocovariance, whereas the exogenous futures moment group omits realized variance (which serves as the target variable).

– **Sentiment** (3 series). Using StockTwits message streams beginning in 2009 for total posts and 2010 for user-tagged posts, we construct three half-hourly indicators: investor attention, defined as the total number of posts within the half hour; a sentiment score that equals 1 when all tagged posts are bullish and $-1$ when all tagged posts are bearish; and the sentiment count, defined as the number of posts carrying either a bullish or bearish tag.

**E30.** Decision: open, for the professor.
Problem: the text gives only the $\pm 1$ endpoints of the sentiment score.
Question for the professor: what is the formula of the score between the endpoints, and what value does it take on intervals with no tagged posts? The repository receives the score precomputed (data/spy_and_sentiment.parquet, column stocktwits_sentiment, first observed in June 2010); src/data/loading.py:280 reads it as a net score in $[-1, 1]$, and no script in src/ or experiments/ computes it.

– **Implied volatility** (3 series). We include three Cboe volatility indices: the 30-day S&P 500 implied volatility index (VIX, available from 2006), the three-month implied volatility index (VIX3M, from 2009), and the implied volatility of the VIX (VVIX, from 2012), all extending through February 2024.

– **Volatility demand** (4 series). Using Cboe option-volume records from 2011 through August 2023, we measure net vega purchased by non-market-maker participants across four series formed by crossing two option universes—options on the S&P 500 index and top S&P 500 exchange-traded funds versus all exchange-traded options—with two transaction subsets, namely all opening and closing customer trades versus opening customer trades only.

– **FOMC** (4 series; used by the pooled intraday model and by the closing-interval specifications of Appendix G). Based on scheduled Federal Open Market Committee announcement timestamps from 1993 through November 2023, we define four series: an indicator for whether an announcement occurs during interval $(t, i)$, an indicator for whether an announcement occurs on day $t$, and two proximity measures $1/(1+n^{+}_{t,i})$ and $1/(1+n^{-}_{t,i})$, where $n^{+}_{t,i}$ and $n^{-}_{t,i}$ denote the number of intervals until the next scheduled announcement and since the most recent scheduled announcement, respectively. Because the meeting schedule is announced in advance, these variables are known prior to the start of the interval.

**E28.** Decision: to decide: disclose the staleness in the text, or add a staleness cap.
Problem: VIX, VIX3M and VVIX end in February 2024, volatility demand in August 2023 and the FOMC series in November 2023, and they are forward-filled with no staleness cap until 30 April 2024, the end of the evaluation span.
Checked on the data: the last observed VIX, VIX3M and VVIX bar is on 12 February 2024 and the last volatility-demand bar on 31 August 2023 (data/vix_and_voldemand.parquet); the last scheduled FOMC print is on 1 November 2023 (data/releases.parquet, column "fomc release"; the loader leaves the tail missing, src/data/loading.py:120–124). The forward fill feeds the trailing baseline statistics, and every interval with $a_{t,i} = 0$ is then stored as the causal median (data cleaning, below), so over the tail the availability averages $\bar{a}^{(k)}$ fall to zero. The text discloses none of this.

– *[name the vendor of each file].*

**E2.** Decision: name the vendor, the price field, the roll, and the bin edges.
Problem: the vendor, the price inside the one-minute log change, the day whose volume picks the front month, and the clock edges of every bin except 15:30–16:00 and the weekend break are missing, so the panel cannot be rebuilt.
The futures moments are now sums of one-minute changes in the log price of the front-month contract, the most active. The text does not say whether that price is a last trade, a midpoint, or a settlement, and it does not name the vendor of the file. It does not say which day's volume makes a contract the most active. The closing bin is 15:30–16:00 ET, and the grid drops Friday 20:00 through Sunday 18:30. Each other bin is a half hour ending on a half-hour stamp. Those stamps are not listed, so the $M = 48$ bins cannot be rebuilt from the text.

- **Intraday grid and data cleaning.**
  - We construct the panel on a 30-minute grid starting from the first observed realized variance. An interval $i$ on day $t$ is retained in the interval grid $\mathcal{G}$ if and only if its realized variance is observed and it falls outside the weekend trading break:

$$(t, i) \in \mathcal{G} \iff \mathrm{RV}_{t,i} \text{ observed and } (t, i) \notin (\text{Fri } 20{:}00,\ \text{Sun } 18{:}30).$$

  - Lag operators step backward sequentially through the retained intervals in $\mathcal{G}$ in chronological order, spanning overnight sessions and trading-day boundaries; thus, $(t, i - j)$ refers to the $j$-th retained interval preceding $(t, i)$, which may lie on an earlier trading day.
  - Exchange holidays and early-close sessions are retained in the futures panel, whereas early-close days are excluded from the option-trading evaluation sample.
  - For each exogenous series $u$, we first record a binary availability (non-missingness) indicator prior to any imputation:

$$a_{t,i} = \mathbf{1}\{u \text{ observed in interval } (t, i)\} \in \{0, 1\}.$$

  - Subsequently, $u_{t,i}$ is forward-filled (carrying the last observed value forward without a staleness cap) for the purpose of computing trailing baseline statistics, after which every interval with $a_{t,i} = 0$ is replaced by the causal median imputation described in step 4 of Section 4.
  - The diurnal adjustment factor $s^2_{t,i}$ and the winsorized target $\tilde{y}_{t,i}$ are formally defined in Section 5.1, with the corresponding rolling window lengths and quantile thresholds reported in Appendix B.1.
- **Estimation and evaluation samples** (see Appendix Table 3).
  - The pooled intraday sample comprises 300,317 half-hour intervals beginning in June 1998, using rolling estimation windows of length $W = 24{,}000$ intervals for the tree ensembles and $W = 96{,}000$ intervals for the linear models (see Appendix B.2):

$$300{,}317 = 303{,}442 \text{ intervals on the grid} - 3{,}125 \text{ burn-in samples}.$$

  - The closing-interval sample ($i = c$, one observation per trading day) used by the closing-interval HAR benchmark, its rolling forecasts and the pre-sample on which the linear penalties are selected are described in Appendix G.1.
  - The primary out-of-sample evaluation span $\mathcal{T}_{\text{oos}}$ covers 3 January 2020 to 30 April 2024 ($T_{\text{oos}} = 1{,}082$ trading days on which all rolling forecasts are available, after excluding 6 early-close sessions). This span begins immediately following the linear models' pre-sample period (Appendix G.1; see Section 5.3).

– The option-trading subsample $\mathcal{T}_{\text{opt}} \subset \mathcal{T}_{\text{oos}}$, on which we evaluate the 0DTE straddle strategy in Section 7, consists of

$$T_{\text{opt}} = 866 \text{ trading days} = 872 \text{ SPXW expirations} - 6 \text{ early-close days}.$$

– Finally, the 419 subsequent SPXW expiration dates (413 full trading days) are held out from the present analysis.

# 4 Predictors

- **Eight groups of exogenous predictors (47 series in total).** As listed in Appendix Table 5, the exogenous series span eight economic categories: realized moments (7), liquidity (14), equal-weighted market moments (6), value-weighted market moments (6), sentiment (3), implied volatility (3), volatility demand (4), and FOMC announcements (4). Section 3 describes the construction of each category, and Appendix Table 5 provides the definition of each individual series:

$$47 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 4.$$

  – Specifically, the four FOMC series are defined as

$$\mathbf{1}\{\text{announcement in } (t,i)\}, \quad \mathbf{1}\{\text{announcement on day } t\}, \quad 1/(1+n^{+}_{t,i}), \quad 1/(1+n^{-}_{t,i}),$$

  – where $n^{\pm}_{t,i}$ denotes the number of intervals until the next $(+)$ and since the last $(-)$ scheduled announcement.

- **Predictors constructed from each series.**
  – Following Corsi (2009), we construct HAR-type trailing averages of each transformed series $\tilde{u}$ over horizons of $k$ samples (one sample is one 30-minute grid interval), lagged by one sample:

$$\bar{u}^{(k)}_{t,i} = \frac{1}{k}\sum_{j=1}^{k} \tilde{u}_{t,i-j}, \qquad k \in \mathcal{K},$$

  where the horizon set is

$$\mathcal{K} = \{1, 5, 25, 125, 625, 3125\} \quad \text{(base 5, 6 horizons)}.$$

  – Here, $k$ counts samples, the retained intervals of the full intraday grid $\mathcal{G}$ (including overnight sessions), in the intraday and the closing-interval model alike. The closing-interval specification uses the same horizon set (Appendix G.2).
  – In addition, we compute the corresponding $k$-interval trailing averages of the availability indicator, denoted $\bar{a}^{(k)}_{t,i}$.
  – We also compute $k$-interval trailing averages of an activity indicator, $\bar{\chi}^{(k)}_{t,i}$, for 6 mode-inflated series in the intraday model.

**E6.** Decision: one rule and one count, the intraday count 6.
Problem: the bullet calls the series mode-inflated, the transformation keeps the series that pass a 20% mode rule, and the intraday column count uses 6.
This bullet calls these series mode-inflated. That word means the raw series has a mode at 30, and 30 is the number of observations: the count of one-minute intervals in a half-hour bar. The transformation further down does not look for zeros. It sets the indicator $\chi_{t,i}$ to 1 when the value differs from the series' full-sample mode, and it keeps a series only when that mode covers at least 20% of intervals. A mode can sit at a number other than zero, so the 20% rule and a mass at zero select different series. The intraday column formula multiplies by 6.
Additional decision (2026-10-17, user): remove the trailing averages of the activity indicators, the $\bar{\chi}_{t,i}^{(k)}$ blocks, from the model.
Note (2026-10-17, user): the count of 7 comes from the closing-interval panel and belongs in the economic application, the section on the last-30-minute closing trade; it may move to the appendix, with a reference back from that section.

  - For notational simplicity in our tables and figures, we suppress the overbar and the interval index $i$ ($i = c$ for the scored forecast), writing $u_t^{(k)}$ for series $u$ averaged over $k$ samples; for instance, $\texttt{vix}_t^{(25)}$ denotes the transformed VIX series averaged over the 25 samples ending at 15:30 on day $t$. We refer to each exogenous series by its column name in the data (Table 5): the implied volatility series are `vix`, `vix3m`, and `vvix`; the suffixes `_ewstock` and `_vwstock` mark the equal- and value-weighted constituent aggregates (e.g., `ofi_ewstock`, `sumret2_vwstock`, and `spread_vwstock`); and the announcement predictors are `fomc_release`, `fomc_day`, `fomc_until_inv`, and `fomc_since_inv`, while writing $\bar{a}_t^{(k)}(u)$ and $\bar{\chi}_t^{(k)}(u)$ for the trailing averages of the availability and activity indicators.

- **HAR benchmark.** The HAR benchmark predictor set consists of the HAR components of the target, $\bar{y}_{t,i}^{(k)}$—constructed via the same trailing-average operator as $\bar{u}_{t,i}^{(k)}$ applied to the winsorized target (6 columns in the intraday model)—together with the calendar vector $\boldsymbol{D}_{t,i}$, which comprises five day-of-the-week dummy variables, the hour of the interval-end timestamp, and the overnight session indicator.

**E9.** Decision: the calendar stops at the overnight indicator, and that indicator stays only if the overnight intervals stay in the training sample (note E15).
Problem: the columns after the overnight indicator are out of the specification, and the printed column counts and Table 1 still include them. Whether the overnight indicator itself stays depends on the sample in note E15.
The sentence above keeps three calendar pieces: the five weekday dummies, the hour of the bar-end stamp, and the overnight indicator. Every column that used to sit after the overnight indicator is out of the specification. That is the opening-interval indicator, the closing-interval indicator, expiration, expiration week, quadruple witching, the quadruple-witching close, month end, quarter end, days to expiration, and the products of the HAR components with the opening and closing indicators. The overnight indicator's own place depends on which sample note E15 keeps. On all 48 intervals it equals 1 off the cash session and again on the 16:00 bar, so it varies across the window and a regression can estimate it. On the 13 regular-hours bars it equals 1 only on the 16:00 bar, so inside that sample it is constant on every bar but the close. The column counts in the formulas below, and the fits in Table 1, were built with the dropped columns still in the matrix.

- **Sequential feature transformations.** Each exogenous series undergoes the following seven transformations in sequence:

  1. *Diurnal adjustment:* we divide each raw series $u_{t,i}$ by a time-of-day scale factor $b_{t,i}^{(u)}$ estimated over the same prior-day set $\mathcal{D}_{t,i}$ as $s_{t,i}^2$ and bounded relative to its own trailing median:

$$\tilde{u}_{t,i} = \frac{u_{t,i}}{b_{t,i}^{(u)}}, \qquad b_{t,i}^{(u)} = \begin{cases} \mathrm{clip}\big(\mathrm{mean}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big), & u \geq 0, \\ \max\big\{\mathrm{clip}\big(\mathrm{sd}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big),\ 0.1\,\bar{\sigma}^{(u)}\big\}, & u \text{ signed}, \end{cases}$$

- where $\rho_{t,i}$ denotes the median of the 250 most recent unbounded diurnal factors for interval $i$ (requiring at least 20 observations),
- $\bar{\sigma}^{(u)}$ denotes the full-sample median of the within-interval standard deviation on the same intraday interval,
- $b_{t,i}^{(u)}$ is set to 1 whenever the factor of a nonnegative series is zero or undefined, and
- no diurnal adjustment is applied to the implied volatility indices (VIX, VIX3M, and VVIX), the StockTwits sentiment series, or the FOMC series.

2. *Variance-stabilizing transformation:* we apply a domain-specific mapping $\tilde{u}_{t,i} \leftarrow g(\tilde{u}_{t,i})$ according to the first matching rule in

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, realized variance, turnover,} \\ & \text{bipower variation, effective spread,} \\ \operatorname{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \operatorname{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log \max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

   - For signed series without a specific power-root rule whose standard-deviation divisor hits its floor on at least 50% of intervals (namely, the volatility-demand series), we instead apply an inverse hyperbolic sine transformation directly to the raw series:

$$\tilde{u}_{t,i} = \operatorname{asinh}(u_{t,i}/\kappa_{t,i}), \qquad \kappa_{t,i} = \max\{e_{t,i-1},\ 0.01 \operatorname{med}_{j\geq 1} e_{t,i-j}\},$$

   - where $e_{t,i}$ denotes an exponentially weighted moving average of $|u|$ computed over nonzero intervals ($u \neq 0$) with a half-life of 12,000 such intervals (nominally 250 days of 48 intervals).

3. *Winsorization:* each series is clipped at the same rolling quantile thresholds used for the target (see Section 5.1), evaluated over the series' own preceding intervals.
4. *Missing-value imputation:* any missing observation is imputed using the expanding median of the series' previously observed transformed values, while retaining the availability indicator $a_{t,i}$ as a separate predictor:

$$\tilde{u}_{t,i} \leftarrow \begin{cases} \tilde{u}_{t,i}, & a_{t,i} = 1, \\ \operatorname{med} \mathcal{A}_{t,i}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} \neq \emptyset, \\ \tilde{u}_{(1)}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} = \emptyset, \end{cases}$$

   - where $\mathcal{A}_{t,i} = \{\tilde{u}_{t',i'} : (t',i') \prec (t,i),\ a_{t',i'} = 1\}$, with $(t',i') \prec (t,i)$ indicating that $(t',i')$ is a retained interval strictly prior to $(t,i)$, and
   - $\tilde{u}_{(1)}$ denotes the transformed value on the series' first observed interval.

5. *Activity indicators:* for the 7 series whose full-sample mode accounts for at least 20% of all intervals, we define a binary activity indicator

$$\chi_{t,i} = \mathbf{1}\{u_{t,i} \neq \operatorname{mode}(u)\}.$$

**S4a.** Decision: the causal rule in src/backtest/executor.py (CAUSAL_MODE_INFLATION, ACTIVE_FROM_OBSERVED; commit b29774a, 15 October 2026) is the specification. Rerun needed under that rule.
Problem: this item describes the old rule, a full-sample mode covering at least 20% of intervals with the indicator read after the forward fill. Under the causal rule the mode is decided from earlier prints only and the flags come from observed prints, so the volatility-demand series are never mode-inflated and the StockTwits sentiment and message-count series are mode-inflated in some years. The printed forecasts (pooled linear models of 6 October, pooled trees of 7 October 2026) predate the rule and use the old one; so does the $\bar{\chi}$ column of Appendix Table 5, whose generator is not re-run (E5). Rewrite this item and regenerate the table after the rerun.

6. *HAR aggregation:* we form the trailing averages $\bar{u}_{t,i}^{(k)}$, $\bar{a}_{t,i}^{(k)}$, $\bar{\chi}_{t,i}^{(k)}$, and $\bar{y}_{t,i}^{(k)}$ across horizons $k \in \mathcal{K}$.
7. *Robust standardization:* under the pooled intraday scheme, we standardize the trailing averages of series values over the rolling estimation window using the median and interquartile range (with lower floors). Specifically, the median and interquartile range are computed exclusively over intervals with $a_{t',i'} = 1$ inside the rolling window $\mathcal{W}_{t,i}$ (see Appendix B.2), re-evaluated every 480 intervals and held constant until the next grid point. Let $M_{t,i}^{(k)}$ denote the maximum such interquartile range observed across all grid points up to $(t, i)$. We apply a 0.01 relative floor in the denominator (note that $\mathrm{IQR}^{*(k)}$, defined in Appendix G.4, does not enter this pooled denominator):

$$z_{t,i}^{(k)} = \begin{cases} \dfrac{\bar{u}_{t,i}^{(k)} - \mathrm{med}\{\bar{u}_{t',i'}^{(k)} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}}{\max\{\mathrm{IQR}_{t,i}^{(k)},\ 0.01\, M_{t,i}^{(k)}\}} & \text{if at least 512 such intervals,} \\ 0 & \text{otherwise,} \end{cases}$$

$$\mathrm{IQR}_{t,i}^{(k)} = \mathrm{IQR}\{\bar{u}_{t',i'}^{(k)} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}.$$

   - On the initial estimation window, both statistics are computed using that window's own rows; an interquartile range below $10^{-12}$ is replaced by 1 on the initial window and, at subsequent grid points, retains the previous scale factor.
   - The indicator averages $\bar{a}^{(k)}$ and $\bar{\chi}^{(k)}$ are left unstandardized.
   - The target HAR components and calendar variables do not use the availability-restricted denominator above; instead, their median and interquartile range are computed across all rows of $\mathcal{W}_{t,i}$ with $\mathrm{IQR}^{*(k)} = 0$ ($\mathrm{IQR}^{*(k)}$ is defined in Appendix G.4), subject to the same 0.01 floor relative to the running maximum interquartile range of that column.

- **Column screening.** At each estimation origin, we screen out constant and duplicate predictors over the active estimation window $\mathcal{W}_t$ by retaining the index set

$$\mathcal{J}_t = \{j :\ \boldsymbol{x}_{\cdot j} \text{ not constant},\ \boldsymbol{x}_{\cdot j} \neq \boldsymbol{x}_{\cdot j'}\ \forall\, j' < j,\ j' \in \mathcal{J}_t\}, \qquad \boldsymbol{x}_{\cdot j} = (x_{t'j})_{t' \in \mathcal{W}_t}.$$

  - Specifically, column equality is evaluated bitwise, and whenever two columns are identical on $\mathcal{W}_t$, the first column in order is retained.

- **Predictor set dimensions.** The column counts of the closing-interval specification are given in Appendix G.2.

  - In the pooled intraday specification, the full predictor set comprises 634 candidate columns, constructed from 47 exogenous series, 6 horizons, and 12 interaction terms

between the target HAR components and the opening- and closing-interval dummy variables. Note that these column counts still reflect the calendar columns and interaction terms discussed in note E9 above:

$$634 = \underbrace{6 + 16}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{12}_{\bar{y}^{(k)} \times \text{open, close}} + 47 \times 6 + 47 \times 6 + 6 \times 6.$$

Here, the 47 intraday exogenous series encompass every category in Section 3, the FOMC group (4 series) included. Appendix Table 5 lists each individual series.

**E35.** Decision: base 5 across the paper, and the pooled model includes the FOMC series (user, 2026-10-19).
Problem: 634 and the formula above now count the base-5 pooled panel with the FOMC series, built on 19 October 2026, before the cuts of notes E6 and E9; no printed forecast was fitted on that panel. The earlier pooled trees and the intraday OLS were fitted on the base-2 panel without the FOMC series, 1,144 columns from 12 horizons and 43 series (note E34). The base-5 panel (experiments/pooled_trees_base5.py, design stage: the repo's own preparation with the data fixes of 15 October 2026, the FOMC calendar fix and the causal mode rule) has 6 horizons, 47 series and 12 interaction columns. Under the causal rule 6 series carry an activity indicator, the number of observations, the three StockTwits series and the FOMC statement-interval and statement-day dummies, where the old full-sample rule gave 5 on the panel without FOMC, the number of observations and the four volatility-demand series (note S4a). With the activity averages dropped (E6) and the calendar cut after the overnight indicator together with the interaction columns (E9), the design has 577 columns on all 48 intervals and 576 on the 13 regular-hours bars, where the overnight indicator is dropped too. Those two counts have no macro yet.

- **Causal construction and absence of look-ahead.** Every rolling statistic indexed by $(t, i)$ is computed strictly from intervals prior to $(t, i)$, ensuring that every predictor of interval $(t, i)$ is in the information set at its forecast origin (15:30 ET for the scored closing interval $i = c$). This includes the HAR trailing averages, the diurnal factor $s^2_{t,i}$, the winsorization quantiles, the median baseline $\rho_{t,i}$, the standardization moments, and the active column set $\mathcal{J}_t$.
  - We disclose three full-sample quantities used in preprocessing: the full-sample mode used to define $\chi_{t,i}$, the within-interval standard-deviation floor $\bar{\sigma}^{(u)}$, and the rule assigning series to the inverse hyperbolic sine transformation.

    **S4b.** Decision: rerun needed under the causal rule (note S4a).
    Problem: the full-sample mode is the old rule, used by the printed forecasts of 6–7 October 2026. Under the causal rule in src/backtest/executor.py the mode is decided from earlier prints only, so it is no longer a full-sample quantity; drop it from this list after the rerun.
  - As noted in Section 3, the FOMC proximity variables rely solely on the pre-announced meeting schedule, which is publicly known ahead of each interval.
  - Any missing interval prior to a series' initial observation is filled with that first observed print (step 4).
  - Finally, standardization statistics on the first estimation window are computed exclusively from burn-in observations.

# 5 Methods

## 5.1 Problem formulation

- **Rolling-window out-of-sample exercise.** We conduct a rolling-window out-of-sample forecasting exercise in which each model is re-estimated every period using only predictors in the information set $\mathcal{F}_t$ available at the 15:30 ET forecast origin. A period corresponds to a 30-minute intraday interval under our pooled intraday scheme (Appendix B.2).

- **Notation.** We index trading days by $t = 1, \ldots, T$ and the 30-minute intraday intervals of day $t$ (including the overnight session) by $i = 1, \ldots, M$, where $i = c$ denotes the closing half-hour interval (15:30–16:00 ET).
  - Realized variance $\mathrm{RV}_{t,i}$ and the lag operator $(t, i-j)$ are defined in Section 3. Our primary forecast target is the realized variance of the closing interval, $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$.
  - We denote by $\mathcal{F}_t$ the information set containing all data observed through 15:30 ET on day $t$, and write $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $\mathrm{Var}_t(\cdot) \equiv \mathrm{Var}(\cdot \mid \mathcal{F}_t)$ for the corresponding conditional expectation and variance.
- **Diurnal adjustment.** Following Andersen and Bollerslev (1997), we remove the intraday diurnal pattern using the trailing sample mean of realized variance in the same time-of-day interval strictly prior to day $t$:

$$s^2_{t,i} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i},$$

$$\mathcal{D}_{t,i} = \{\text{last } n_{\mathrm{d}} \text{ trading days } t' < t \text{ with interval } i\}, \qquad |\mathcal{D}_{t,i}| \geq \underline{n}_{\mathrm{d}}.$$

  - For intervals with $|\mathcal{D}_{t,i}| < \underline{n}_{\mathrm{d}}$, we set $s^2_{t,i} = 1$ and exclude the interval from evaluation. The window parameters $n_{\mathrm{d}}$ and $\underline{n}_{\mathrm{d}}$ are given in Appendix B.1.
- **Regression target.** To stabilize the variance, we take the square root of the diurnally adjusted realized variance and winsorize the resulting series at rolling lower and upper empirical quantiles:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s^2_{t,i}}, \qquad \mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, n_{\mathrm{w}}\},$$

$$\tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ \underline{Q}_{t,i},\ \overline{Q}_{t,i}).$$

  - Here $\underline{Q}_{t,i}$ and $\overline{Q}_{t,i}$ denote the lower and upper empirical quantiles computed over $\mathcal{Y}_{t,i}$, which pools the preceding $n_{\mathrm{w}}$ intervals across all times of day, and clip truncates its first argument to that range. The quantile levels and window length $n_{\mathrm{w}}$ are reported in Appendix B.1.
  - All models are estimated on the winsorized target $\tilde{y}_{t,i}$, whereas out-of-sample evaluation is conducted against the unwinsorized realized variance $\mathrm{RV}_{t,c}$.
  - For the scored interval $i = c$ (the closing half hour that every model forecasts), we write $y_t \equiv y_{t,c}$ and $\tilde{y}_t \equiv \tilde{y}_{t,c}$.
- **Back-transformation.** To obtain a variance forecast in raw units, we square the fitted volatility and multiply by the diurnal factor without post-estimation recalibration:

$$\hat{\sigma}^2_t = \hat{y}_t^{\,2}\, s^2_{t,c},$$

  where $\hat{y}_t \equiv \hat{y}_{t,c}$ is the forecast of the winsorized target $\tilde{y}_t$, not of the unclipped $y_t$, formed using $\mathcal{F}_t$.
  - Because $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t[y_t])^2 = \mathrm{Var}_t(y_t) \geq 0$, squaring the level forecast introduces a downward Jensen inequality bias that we leave uncorrected (note E33).
- **Rolling estimation window.** Under the intraday scheme, the estimation window at origin $(t, i)$ is $\mathcal{W}_{t,i}$ (Appendix B.2), consisting of the $W$ most recent intervals prior to $(t, i)$. We let $\mathcal{I}$ denote the subset of intraday intervals included in estimation.

- **Hyperparameter-selection splits.** In line with Gu et al. (2020), we partition the rolling window forward in time into an estimation subsample, an embargo gap of $n_{\text{gap}}$ observations, and a validation subsample of $n_{\text{val}}$ observations:

$$\mathcal{W}_{t,i}^{\text{est}} = \{(t,\, i - W), \ldots, (t,\, i - n_{\text{val}} - n_{\text{gap}} - 1)\},$$
$$\mathcal{W}_{t,i}^{\text{gap}} = \{(t,\, i - n_{\text{val}} - n_{\text{gap}}), \ldots, (t,\, i - n_{\text{val}} - 1)\},$$
$$\mathcal{W}_{t,i}^{\text{val}} = \{(t,\, i - n_{\text{val}}), \ldots, (t,\, i - 1)\}.$$

  Target interval $(t, i)$ never enters any of these sets. The lengths $n_{\text{gap}}$ and $n_{\text{val}}$ are given in Appendix B.1.
- **Estimation objective.** Each model is estimated by minimizing the sum of squared errors on the scale of $\tilde{y}$. At forecast origin $(t, i)$, the parameter vector $\boldsymbol{\theta}$ is fitted over the full rolling window $\mathcal{W}_{t,i}$:

$$\hat{\boldsymbol{\theta}}_{t,i} = \arg\min_{\boldsymbol{\theta}} \sum_{(t',i') \in \mathcal{W}_{t,i}} \left(\tilde{y}_{t',i'} - \hat{y}_{t',i'}(\boldsymbol{\theta})\right)^2.$$

  - Here $(t', i')$ indexes the historical intervals in the window $\mathcal{W}_{t,i}$, and $\boldsymbol{\theta}$ collects the model parameters.
  - During hyperparameter tuning only, candidate configurations are estimated on $\mathcal{W}_{t,i}^{\text{est}}$ rather than $\mathcal{W}_{t,i}$ and evaluated on $\mathcal{W}_{t,i}^{\text{val}}$.

## 5.2 Models

- **HAR and the HAR benchmark via OLS.** Following Corsi (2009), we specify the linear heterogeneous autoregressive model for each intraday interval $(t, i)$ as

$$\hat{y}_{t,i} = \beta_0 + \sum_{k \in \mathcal{K}} \beta^{(k)}\, \bar{y}_{t,i}^{(k)} + \boldsymbol{\delta}^\top \boldsymbol{D}_{t,i}.$$

  - The horizon set $\mathcal{K}$ and the lagged volatility averages $\bar{y}_{t,i}^{(k)}$ are defined in Section 4, where $\bar{y}_{t,i}^{(k)}$ is the mean of the winsorized target over the $k$ half-hour intervals preceding $(t, i)$, including overnight intervals. The scored forecast is the closing-interval prediction $\hat{y}_t \equiv \hat{y}_{t,c}$.
  - The vector $\boldsymbol{D}_{t,i}$ collects the calendar dummy variables with coefficient vector $\boldsymbol{\delta}$; a pure HAR specification restricts $\boldsymbol{\delta} = \mathbf{0}$. Our HAR benchmark ($\mathcal{X} = \text{HAR}$) includes both the HAR components and the calendar dummies.
  - The full predictor vector $\boldsymbol{x}_{t,i} \in \mathbb{R}^p$ ($\mathcal{X} = \text{Full}$) augments the HAR benchmark with the multi-horizon averages $\bar{u}_{t,i}^{(k)}$ ($k \in \mathcal{K}$) and the corresponding availability and activity indicators for every exogenous series $u$ (Section 4).
- **Penalized linear regressions: ridge (Hoerl and Kennard, 1970), lasso (Tibshirani, 1996), and elastic net (Zou and Hastie, 2005).** To accommodate the high-dimensional predictor vector $\boldsymbol{x}_{t,i}$, we augment the residual sum of squares with an $\ell_1$, $\ell_2$, or combined penalty, following the penalized HAR formulation of Zhang et al. (2024), and update the coefficient estimates as the estimation window rolls forward:

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{ridge}} = \arg\min_{\beta_0, \boldsymbol{\beta}} \sum_{(t',i') \in \mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{enet}} = \arg\min_{\beta_0, \boldsymbol{\beta}} \sum_{(t',i') \in \mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda\alpha \sum_{j=1}^{p} \omega_j |\beta_j| + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

where $\lambda > 0$ governs overall penalty strength, $\alpha \in [0, 1]$ is the elastic-net mixing parameter, and $\omega_j \geq 0$ is the predictor-specific penalty weight (with the intercept $\beta_0$ unpenalized, $\omega_0 = 0$). For ridge, $\lambda$ coincides directly with the grid value reported in Table 4. For lasso and elastic net, Table 4 reports the `scikit-learn` penalty parameter, so the penalty multiplier $\lambda$ in the unnormalized objective above equals $2n$ times the tabulated value, where $n$ is the number of observations in the window.

  - **Differential block penalty.** When estimating ridge, lasso, and elastic net on the full predictor set, we penalize the HAR benchmark predictors 100 times more lightly than the exogenous predictors:

$$\omega_j = \begin{cases} \underline{\omega} = 0.01 & j \in \text{HAR benchmark}, \\ 1 & j \text{ an additional predictor}, \\ 0 & j = 0 \text{ (intercept)}. \end{cases}$$

  - When a penalized regression is fitted on the HAR benchmark alone, we apply a uniform penalty weight $\omega_j = 1$ across all predictors.
  - The lasso corresponds to $\alpha = 1$, and our elastic net specification fixes $\alpha = 0.5$.
  - For ridge regression, we update $(\boldsymbol{X}_{t,i}^\top \boldsymbol{X}_{t,i} + \lambda \operatorname{diag}(\boldsymbol{\omega}))^{-1}$ via exact rank-one downdates and updates as observations exit and enter $\mathcal{W}_{t,i}$, where $\boldsymbol{X}_{t,i}$ stacks the rows $(1, \boldsymbol{x}_{t',i'}^\top)$ for $(t', i') \in \mathcal{W}_{t,i}$ and $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_p)^\top$.
  - For the lasso and elastic net, we update the active set and coefficient vector via exact online homotopy (Garrigues and El Ghaoui, 2008), adding the incoming observation and dropping the outgoing observation at each step.
  - At every estimation step, only the screened predictor subset $\mathcal{J}_{t,i}$ enters the solver: columns that are constant within $\mathcal{W}_{t,i}$ or exact duplicates of an earlier retained column are omitted. In models estimated solely on the closing interval, this screen automatically removes the hour-of-day and overnight indicators (Section 4).

- **Tree ensembles.** To capture potential nonlinearities and high-order interactions among predictors, we consider two gradient-boosted tree implementations (LightGBM and XGBoost) and random forests.

  - **Gradient boosting** (Friedman, 2001), implemented via LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016), minimizes the squared-error loss $\frac{1}{2}(\tilde{y} - F)^2$ over boosting rounds $b = 1, \ldots, B$ with learning rate $\eta$. Each regression tree $f_b$ partitions the predictor space into $J$ leaves with leaf weights $\boldsymbol{w}$, where leaf $j$ contains the observation subset $\mathcal{R}_j$ of size $n_j = |\mathcal{R}_j|$. When the $\ell_1$ leaf penalty $\lambda_1^{\text{tree}} = 0$, the additive update and optimal leaf weight take the form

$$F_b = F_{b-1} + \eta f_b,$$
$$w_j^* = -\frac{\sum_{(t',i') \in \mathcal{R}_j} \big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big)}{n_j + \lambda_2^{\text{tree}}}.$$

    At round $b$, tree $f_b$ is grown to minimize the regularized second-order objective

$$\sum_{(t',i') \in \mathcal{W}_{t,i}} \Big[ \big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big) f_b(\boldsymbol{x}_{t',i'}) + \tfrac{1}{2} f_b(\boldsymbol{x}_{t',i'})^2 \Big] + \gamma^{\text{tree}} J + \lambda_1^{\text{tree}} \|\boldsymbol{w}\|_1 + \tfrac{1}{2} \lambda_2^{\text{tree}} \|\boldsymbol{w}\|_2^2,$$

    where $\gamma^{\text{tree}}$ penalizes tree complexity per leaf and $\lambda_1^{\text{tree}}, \lambda_2^{\text{tree}}$ apply $\ell_1$ and $\ell_2$ shrinkage to the leaf weights $\boldsymbol{w}$.

– Every leaf is required to contain at least $n_{\min}$ observations ($n_j \geq n_{\min}$), which corresponds to the minimum leaf size in LightGBM and, under unit Hessian for squared-error loss, the minimum child weight in XGBoost.
– **Random forest** (Breiman, 2001) forms the equally weighted average of $B$ regression trees $f_b$, each grown independently on a bootstrap sample of the estimation window with a random subset of candidate predictors evaluated at each split:

$$\hat{y}_{t,i} = \frac{1}{B} \sum_{b=1}^{B} f_b(\boldsymbol{x}_{t,i}).$$

### 5.3 Forecast evaluation

- **Forecast and realization.** We evaluate the unadjusted back-transformed variance forecast $\hat{\sigma}_t^2$ defined in Section 5.1 against the actual, unwinsorized realized variance of the closing interval, $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$ (winsorization is applied to $\tilde{y}_{t,c}$ during model estimation only).
- **Evaluation set.** Because every forecast is generated by a model estimated on a strictly trailing window, all predictions are out of sample. We let $\mathcal{T}_{\mathrm{oos}}$ denote the set of common forecast-origin trading days following the initial estimation window and hyperparameter presample on which all competing forecasts are available, excluding early-close days, and write $T_{\mathrm{oos}} = |\mathcal{T}_{\mathrm{oos}}|$ (the exact sample dates and $T_{\mathrm{oos}}$ are given in Section 6.1).
- **Primary loss criterion.** Our primary statistical loss is the daily quasi-likelihood (QLIKE) loss of Patton (2011), which is robust to measurement noise in the realized-variance proxy. Because the diurnal factor $s_{t,c}^2$ cancels between $\mathrm{RV}_t = y_t^2 s_{t,c}^2$ (with unwinsorized $y_t$) and $\hat{\sigma}_t^2 = \hat{y}_t^2 s_{t,c}^2$, the daily loss takes the form

$$L_t = \frac{y_t^2}{\hat{y}_t^2} - \log \frac{y_t^2}{\hat{y}_t^2} - 1 = \mathrm{QLIKE}(\mathrm{RV}_t,\, \hat{\sigma}_t^2),$$

  with sample average over $\mathcal{T}_{\mathrm{oos}}$ given by

$$\overline{\mathrm{QLIKE}} = \bar{L} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} L_t.$$

- **Benchmark forecast.** Our baseline is the closing-interval HAR benchmark: a HAR-type model in the spirit of Corsi (2009), estimated by OLS on the trailing averages $\bar{y}^{(k)}$ of the target over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples and the 16 calendar columns $\boldsymbol{D}$ (Section 5.2), using closing-interval observations only (Appendix G.3).
- **Relative loss.** When comparing a candidate forecast $A$ against a reference forecast $B$ (the closing-interval HAR benchmark unless noted otherwise), we report the percentage difference in mean QLIKE:

$$\Delta\% = 100\Big(\frac{\bar{L}^A}{\bar{L}^B} - 1\Big),$$

  and assess its statistical significance via the Diebold–Mariano test detailed in Appendix I.
- **Squared-error loss.** In variance units, we also compute the mean squared error over $\mathcal{T}_{\mathrm{oos}}$:

$$\mathrm{MSE} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \big(\mathrm{RV}_t - \hat{\sigma}_t^2\big)^2.$$

- **Out-of-sample $R^2$.** Following Campbell and Thompson (2008), we compute $R^2_{\text{OOS}}$ relative to the expanding historical mean of closing-interval realized variance known at the forecast origin:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t\in\mathcal{T}_{\text{oos}}}\left(\text{RV}_t - \hat{\sigma}^2_t\right)^2}{\sum_{t\in\mathcal{T}_{\text{oos}}}\left(\text{RV}_t - \overline{\text{RV}}_{t-1}\right)^2}, \qquad \overline{\text{RV}}_{t-1} = \frac{1}{|\{t' < t\}|}\sum_{t'<t}\text{RV}_{t'},$$

where the expanding average $\overline{\text{RV}}_{t-1}$ is taken over all prior trading days from the sample start specified in Section 6.1.

**E33.** Decision: to decide: compute MAE and the calibration diagnostics (the ratio $\overline{\text{RV}}/\overline{\hat{\sigma}^2}$ and the under-prediction frequency) for the forecasts of Table 1, in make_rehaul_oos_table.py, or drop them.
Problem: MAE and calibration diagnostics promised here are not yet computed/reported.
No generator of the paper (writeup/make_rehaul_*.py) computes MAE, the ratio or the under-prediction frequency. The calibration files under results/atm_straddle_0dte_1530/ (recal_grid/, recal_2026-09-24/) come from the earlier recalibration studies, not from the current forecasts, and generated/campaign_numbers.tex holds one MAE of an earlier campaign, which this paper does not load.

# 6 Experiments and results

## 6.1 Experimental setup

- **Evaluation span $\mathcal{T}_{\text{oos}}$** (see Section 5.3). We evaluate the rolling out-of-sample forecasts over the period from 3 January 2020 to 30 April 2024, which comprises $T_{\text{oos}} = |\mathcal{T}_{\text{oos}}| = 1{,}082$ trading days on which all competing forecasts are available (after excluding 6 early-close sessions). This evaluation span begins with the pooled intraday linear forecasts immediately following the pre-sample validation period (6 January 2016 to 2 January 2020) used to select the linear regularization penalties.
- **Training scheme.** Under the pooled intraday estimation scheme, models are fit across all $M = 48$ half-hour intervals of each trading day using a rolling window of $W = 24{,}000$ intervals for the tree ensembles and $W = 96{,}000$ intervals for the linear specifications. On the full predictor set, the pooled LightGBM and XGBoost forecasts reflect an earlier specification and the pooled random forest forecast is preliminary (see Table 1).

**E15.** Decision: one hyperparameter rule for every class, frozen after a single pre-sample pick.
Problem: the sample and the window are still open, and the printed QLIKE cells for the linear light re-tune, the HAR-benchmark $\lambda$ path, the full-set LightGBM draft, the full-set XGBoost draft, and the preliminary random forest were not produced by one frozen configuration.
The rule for the comparison is one configuration per model, chosen once on the pre-sample and then held fixed for the whole evaluation span. Two inputs to that rule are still unset, and the only write-up of the comparison is in the author notes. The sample is either every one of the 48 intervals or the 13 regular-hours bars whose end stamps run from 10:00 to 16:00. The window is either $W = 24{,}000$ intervals or $W = 96{,}000$ intervals. On top of that open choice, some cells already printed in Table 1 were not produced by one frozen configuration. The linear penalties still move under the light re-tune, the HAR-benchmark penalties move on the dates left open in note E23, and the full-set LightGBM row, the full-set XGBoost row, and the random-forest row are an earlier draft or a preliminary fit. Those QLIKE numbers are not the losses of one shared configuration, and the table has not been refit.

- **Benchmark forecast.** Our baseline benchmark is the closing-interval HAR benchmark, a HAR-type model in the spirit of Corsi (2009) on the target's trailing averages over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples and the 16 calendar columns, estimated by OLS on closing-interval observations (Appendix G.3); Appendix G.5 examines all model classes trained analogously on the closing 30-minute interval alone.
- **Rolling re-estimation and causal tuning.** The pooled intraday models are re-estimated at every 30-minute interval (see Appendix B.2), with hyperparameters either held fixed or

selected causally on an initial pre-sample or trailing validation windows prior to the forecast origin.

- **Evaluation metric and inference.** We evaluate predictive accuracy via the QLIKE loss against actual realized variance (see Section 5.3) and conduct pairwise inference using Diebold–Mariano tests with the Harvey–Leybourne–Newbold small-sample adjustment and a Newey–West long-run variance estimator ($q_{\mathrm{NW}} = 6$ on $\mathcal{T}_{\mathrm{oos}}$).[1]

We refer the reader to Appendix B for comprehensive implementation details.

## 6.2 Out-of-sample accuracy

- **Overview of forecasting performance.** Table 1 reports the out-of-sample MSE, average QLIKE loss, and out-of-sample $R^2_{\mathrm{OOS}}$ for each pooled intraday model class under both the HAR benchmark and the full predictor set across the 1,082 trading days of the evaluation span $\mathcal{T}_{\mathrm{oos}}$. The corresponding metrics for specifications estimated exclusively on the closing half-hour interval—including the OLS regression on the HAR benchmark predictor set, our closing-interval HAR benchmark forecast—are presented in Table 2 (see Appendix G.5). Split at the start of 2020, the full predictor set has the lower QLIKE in both parts (2012–2019: 0.1718–0.1740 against 0.1840 for the HAR benchmark; 2020–2024: 0.1869–0.1874 against 0.1951–0.1952), and a higher $R^2_{\mathrm{OOS}}$ on 2012–2019 (68.8–69.3% against 66.4%) but a lower one on 2020–2024 (38.1–45.7% against 66.4%), so its lower $R^2_{\mathrm{OOS}}$ over the whole span comes from 2020–2024.

  **E36.** Decision (user, 2026-10-20): Table 1 is like for like. It holds the pooled linear rerun trained and scored on the 13 regular-hours bars (base-5 HAR horizons, FOMC series in, a rolling window of 1,000 trading days, penalties chosen once on four validation folds covering 2008–2011 and then frozen), with every bar scored from 3 January 2012 to 30 April 2024 (3,075 trading days, one panel). Its tests are Diebold–Mariano on the daily average QLIKE. The 2,000-day reruns (from 2020 and from 2012) and the 48-interval rerun are not in the table; their numbers are in results/oos_rerun_2026-10-20/ and results/oos_span_check_2026-10-20/. The rest of this section, the abstract, the introduction and the conclusion still describe the span from 3 January 2020; results/oos_span_check_2026-10-20/TABLE1_SWITCH.md proposes the changes. Tree ensembles are held off until they are refit on the cluster.
  Problem: this bullet and the $R^2_{\mathrm{OOS}}$ bullet below still describe the earlier table, and so do the abstract, the introduction and the conclusion. They speak of the 15:30 forecast of the closing half hour, six model classes with trees, a closing-half-hour $R^2$ benchmark, and the `\rhrAfPool...` counts that writeup/make_rehaul_results.py computed from the earlier runs.
  New tests, QLIKE, all features against the HAR benchmark of the same model, $p$-values in parentheses: ridge $-7.86$ ($<0.001$), lasso $-8.07$ ($<0.001$), elastic net $-10.45$ ($<0.001$). The like-for-like values of the prose numbers are being computed (branch insurance-selling-2026-09-19-prose-numbers).

- **Computation of $R^2_{\mathrm{OOS}}$.** Following Campbell and Thompson (2008), the reference forecast in $R^2_{\mathrm{OOS}}$ is the expanding historical mean of closing-half-hour realized variance computed from 5 January 1998 through day $t-1$, which relies solely on information available prior to the forecast origin.

## 6.3 The full predictor set versus the HAR benchmark

- **Paired comparison within each model class.** To isolate the predictive contribution of the exogenous features within each intraday model class $m$ ($\mathcal{I} = \{1, \ldots, M\}$), we form the daily loss differential $d_t = L_t^{\mathrm{Full}} - L_t^{\mathrm{HAR}}$, where $L_t^{\mathcal{X}}$ denotes the QLIKE loss of the forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$. We report the percentage change in average loss $\Delta\%$ defined in Section 5.3,

[1] Appendix I provides the complete specification of these statistical tests.

along with the Diebold–Mariano statistic DM computed on $d_t$ over $\mathcal{T}_{\text{oos}}$ using lag truncation $q_{\text{NW}} = 6$.

- **Consistent QLIKE reductions across all model classes.** Augmenting the HAR benchmark with the full predictor set reduces the average out-of-sample QLIKE loss across every intraday model class ($\Delta\% < 0$ in all 6 classes), with 4 of these reductions statistically significant at the 5% level (see Table 1):
  - Regarding the tree ensembles, we observe the largest relative improvements: $\Delta\% = -13.0$ for LightGBM (DM $= -4.39$), $-12.0$ for XGBoost (DM $= -4.29$), and $-11.9$ for the random forest (DM $= -4.75$), each of which is statistically significant at the 5% level.
  - Regarding the penalized linear models, the elastic net achieves a statistically significant reduction of $\Delta\% = -5.6$ (DM $= -2.10$), while ridge regression and the lasso yield reductions of $-5.0$ (DM $= -1.77$, $p = 0.077$) and $-3.0$ (DM $= -1.13$, $p = 0.26$), respectively.

### 6.4 Model classes

- **Exogenous features and the predictability of realized variance, by model class.** Comparing across model classes in Table 1, the penalized linear models consistently achieve lower average QLIKE loss than the tree ensembles on both predictor sets:
  - Under the full predictor set, the linear models attain average QLIKE losses between 0.1078 and 0.1101—specifically, 0.1078 for the elastic net, 0.1086 for ridge, and 0.1101 for the lasso, representing the three most accurate specifications among all 12 pooled intraday forecasts—compared with 0.1126–0.1146 for the tree ensembles.
  - Under the HAR benchmark, the linear specifications achieve average QLIKE losses of 0.1135–0.1143, outperforming the tree ensembles at 0.1279–0.1301.
- **Comparison against the closing-interval HAR benchmark.** Every pooled intraday specification incurs a higher average QLIKE loss than the closing-interval HAR benchmark ($\overline{\text{QLIKE}} = 0.1033$):
  - Under the full predictor set, the average QLIKE loss of the pooled linear models exceeds that of the closing-interval HAR benchmark by +4.4 to +6.7%, although none of these differences is statistically significant at the 5% level; for the pooled tree ensembles, the excess loss ranges from +9.0 to +11.0%, each significant at the 5% level.
  - Under the HAR benchmark, the excess QLIKE loss of the pooled intraday models ranges from +10.0 to +26.0%, with every difference statistically significant at the 1% level.
  - Consequently, incorporating the exogenous covariates narrows the accuracy gap relative to the closing-interval HAR benchmark across every intraday model class.
  - Because the closing-interval HAR benchmark is estimated solely on the closing 30-minute interval of each day, we extend this closing-interval training scheme to all model classes in Appendix G.5.

### 6.5 Variable importance

Variable importance is not yet computed for the pooled intraday model; the closing-interval analysis is parked (note E19).

**Table 1:** Forecast accuracy of the intraday models, 3 January 2012 to 30 April 2024 (1,000-day window, penalties frozen from 2008–2011)

| | HAR benchmark | | | All features | | | |
|---|---|---|---|---|---|---|---|
| Model | MSE ($10^{-10}$) | QLIKE | $R^2_{\text{OOS}}$ (%) | MSE ($10^{-10}$) | QLIKE | $R^2_{\text{OOS}}$ (%) | DM |
| OLS | 0.94 | 0.1879 | 66.4 | – | – | – | – |
| Ridge | 0.94 | 0.1879 | 66.4 | 1.52 | 0.1774 | 45.4 | −7.86 (<0.001) |
| Lasso | 0.94 | 0.1879 | 66.4 | 1.58 | 0.1773 | 43.4 | −8.07 (<0.001) |
| Elastic net | 0.94 | 0.1879 | 66.4 | 1.40 | 0.1787 | 49.7 | −10.45 (<0.001) |

*Notes:* Pooled intraday linear models, one design for every row: the HAR components at horizons $k \in \{1, 5, 25, 125, 625, 3125\}$, the full predictor set with the FOMC series, penalties chosen once on four validation folds covering 2008–2011, before the first forecast, and then frozen, a rolling window of 1,000 trading days, re-estimated every interval (Section 5), trained and scored on the 13 regular-hours half-hour bars ending 10:00 to 16:00 ET. Every bar is scored on the $T_{\text{oos}} = 3{,}075$ trading days from 3 January 2012 to 30 April 2024 (early closes dropped), 39,975 bars (bars with a non-positive realized variance dropped); each forecast $\hat{\sigma}^2_{t,i} = \hat{y}^2_{t,i}\, s^2_{t,i}$ is scored against the actual realized variance $\text{RV}_{t,i}$ of its interval. MSE (in units of $10^{-10}$) and QLIKE are averages over the scored intervals. $R^2_{\text{OOS}} = 1 - \sum(\text{RV}_{t,i} - \hat{\sigma}^2_{t,i})^2 / \sum(\text{RV}_{t,i} - \overline{\text{RV}}_{t-1,i})^2$, in percent, where $\overline{\text{RV}}_{t-1,i}$ is the historical mean realized variance of the same half-hour interval over every earlier trading day from 5 January 1998 (Campbell and Thompson, 2008). DM: the Diebold–Mariano statistic, with the correction of Harvey et al. (1997), for the daily average QLIKE of the full predictor set minus that of the HAR benchmark of the same model (negative: the full set has the lower loss), Newey–West lag 8, $p$-value in parentheses. Tree ensembles are left out until they are re-estimated on this design. Models trained on the last 30 minutes are in Table 2.

**E19.** Decision (user, 2026-10-17): compute feature importance for the pooled intraday model with a monthly refit cadence, the default; the user has not chosen between weekly and monthly.
Problem: the pooled intraday model has no feature-importance analysis, and the parked closing-interval analysis refit every 10 trading days (writeup/make_rehaul_fi.py:446).
User decision 2026-10-17: switch to a monthly feature-importance refit; requires a run of feature importance for the pooled intraday model. No monthly cadence has been run, so this section reports none.

# 7 Economic value of the forecasts

## 7.1 The straddle strategy

- **Position.** We trade same-day-expiring (0DTE) SPXW options. With $S_t$ denoting the underlying print on the option chain at the 15:30 ET forecast origin, we select the call and put at the nearest adjacent strikes with live quotes:

$$K_C = \min\{K \geq S_t\}, \qquad K_P = \max\{K \leq S_t\}.$$

  When $K_C = K_P = S_t$, the package forms an at-the-money straddle; otherwise, it forms a narrow strangle on the two strikes bracketing $S_t$. We refer to both positions as the straddle: an approximately delta-neutral package entered at 15:30 ET and held unhedged until expiration at 16:00 ET.

**E18.** Decision: write the day filter into the text, and give the implied-volatility line the recovery used on the censored days.
Problem: the day-drop rule (the wider strike more than 10 index points from $S_t$, or fewer than 10 quoted contracts) is still only a comment under this item, and the implied-volatility line and Appendix C make a censored leg missing, while the scored days recover it.
$S_t$ is now the 15:30 underlying print on the option chain and $S_t^{\text{close}}$ the official S&P 500 close. The notebook checks that the Yahoo closes match the settlement column on every scored day. The chain field is copied as printed, and the code never labels it cash or futures, so the text calls it neither. A day is dropped when the wider strike is more than 10 index points from $S_t$, or when fewer than 10 contracts are quoted. That rule is in Appendix C and is still only a comment under this item. Early-close days are a separate drop, already in the $T_{\text{opt}}$ line of the Data section. The vendor field is an hourly standard deviation from a bisection on [0.0005, 0.025]. Dividing by $\sqrt{2}$ is the half-hour conversion, and that is what was scored. On the 5 scored days where a leg is a failed-solve node, the hourly volatility is recovered by inverting the Black–76 price of the straddle on the 15:30 underlying, then divided by $\sqrt{2}$. Appendix C says a censored leg makes the implied variance missing. That is the drop path, not the scored path.

- **Entry prices.** From the legs' bid quotes $b$ and ask quotes $a$ at 15:30 ET, we define the package mid-quote $P_t$, bid $P_t^b$, and ask $P_t^a$ as

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement.** The package settles in cash at the official S&P 500 close $S_t^{\text{close}}$ with no intraday delta hedging; the simple return on the mid-quote premium paid at entry is

$$X_t = (S_t^{\text{close}} - K_C)^+ + (K_P - S_t^{\text{close}})^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

- **Implied variance.** Let $\text{IV}_t^C$ and $\text{IV}_t^P$ denote the call and put implied volatilities at 15:30 ET, quoted on a per-hour basis. The implied volatility over the closing half hour is

$$\text{IV}_t = \frac{\text{IV}_t^C + \text{IV}_t^P}{2\sqrt{2}},$$

  and $\text{IV}_t^2$ is the corresponding implied variance to expiration.
- **Forecast variance risk premium.** Following Carr and Wu (2009) and Bollerslev et al. (2009), the variance risk premium over the closing half hour is $\text{IV}_t^2 - \mathbb{E}_t[\text{RV}_t]$, which we forecast by $\text{IV}_t^2 - \hat{\sigma}_t^2$, where $\hat{\sigma}_t^2$ is the model variance forecast from Section 5.1. Every trading rule below conditions on the forecast exclusively through the variance gap $\hat{\sigma}_t^2 - \text{IV}_t^2$, taking a long straddle position when forecast variance exceeds implied variance.
- **Trading strategies.** A strategy chooses a daily straddle position $q_t$ ($q_t > 0$ long, $q_t < 0$ short) and earns the daily return $r_t = q_t R_t$. Because the variance forecast enters solely through $q_t$, comparing performance across rules isolates the economic value of the forecast (Fleming et al., 2001). We consider four rules:
  - *Unconditional short-straddle benchmark:* a static short-variance position setting $q_t \equiv -1$;
  - *Sign strategy:* $q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2)$, with ties mapped to $q_t = -1$;
  - *Robust clip:* $q_t = \text{clip}((\hat{\sigma}_t^2 - \text{IV}_t^2)/\text{MAD}_t, -1, 1)$, where $\text{MAD}_t = \text{median}_{t'<t}|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2|$ is the expanding median absolute variance gap over earlier days $t' < t$ in $\mathcal{T}_{\text{opt}}$;
  - *Percentile rank:* the sign of the gap scaled by the empirical percentile rank of its magnitude across the $t-1$ preceding days $t' < t$ in $\mathcal{T}_{\text{opt}}$:

$$q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2) \cdot \frac{1}{t-1} \sum_{t'<t} \mathbf{1}\{|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2| \leq |\hat{\sigma}_t^2 - \text{IV}_t^2|\}.$$

- **Sizing sample.** Every position $q_t$ is constructed causally using days $t' < t$ only. We reserve the first 252 days of $\mathcal{T}_{\text{opt}}$ to initialize the scaling factor $\text{MAD}_t$, and evaluate all trading rules—including the sign strategy—on the remaining 614 post-warmup trading days ($t > 252$).
- **Execution.** At mid-quotes, the strategy return is $r_t = q_t R_t$. At quoted bid and ask prices, buying the straddle pays the ask $P_t^a$ and selling receives the bid $P_t^b$; because settlement occurs in cash at expiration, the bid–ask spread is incurred at entry only:

$$r_t^{\text{ba}} = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- **Transaction costs.** Executing at the quoted bid or ask rather than the mid-quote imposes a median half-spread cost of 2.9% of the straddle premium. No additional commissions or exchange fees are deducted.

### 7.2 Performance measures

- **Annualized Sharpe ratio.** We compute the annualized Sharpe ratio from daily strategy returns, both at mid-quotes and at quoted bid/ask prices, over the 614 post-warmup days of $\mathcal{T}_{\text{opt}}$ ($t > 252$):

$$\text{SR} = \sum_y w_y \,\text{SR}_y, \qquad \text{SR}_y = \sqrt{N_y}\,\frac{\bar{r}_y}{\text{sd}_y(r)}, \qquad w_y = \frac{N_y}{\sum_{y'} N_{y'}},$$

  where $y$ runs over the calendar years of those 614 days, $N_y$ is the number of them that fall in year $y$ (option days, i.e. days with a same-day SPXW expiration), and $\bar{r}_y$ and $\text{sd}_y(r)$ are the sample mean and standard deviation (divisor $N_y - 1$) of $r_t$ over those $N_y$ days.

  **N15b.** Decision: to decide (rerun needed).
  Problem: the interval-shift numbers (`shiftSwingMed*`, `shiftNoiseBand` $= 1.96\sqrt{252/n}$; Section 7.3 and the appendix shift-test paragraph) come from experiments/close_shift_test_headline.py, which still annualizes with the fixed $\sqrt{252}$ on all option days. They were not regenerated: that script rewrites results/close_shift_test/ and its figures. The compounding growth rate $G$ below keeps its factor 252: it annualizes a mean log growth rate, not a Sharpe ratio.

  The 216 days of $\mathcal{T}_{\text{oos}}$ without a same-day SPXW expiration enter the statistical accuracy evaluation only. Because the position is held for 30 minutes intraday, the risk-free return over the holding period is negligible and omitted.
- **Mean return and HAC $t$-statistic.** We test the statistical significance of mean daily returns using the Newey–West HAC $t$-statistic of Appendix I applied to $r_t$ over $|\mathcal{O}| = T_{\text{opt}}$ days with lag truncation $q_{\text{NW}} = 14$.
- **Hit rate and position counts.** We also record the empirical hit rate $\frac{1}{T_{\text{opt}}} \sum_{t \in \mathcal{T}_{\text{opt}}} \mathbf{1}\{r_t > 0\}$ and the number of long-straddle days $\#\{t : q_t > 0\}$.
- **Gain over the short-straddle benchmark.** For any strategy $A$ compared against a benchmark strategy $B$ (the unconditional short straddle unless noted otherwise), we define the daily return differential $\Delta_t = r_t^A - r_t^B$ and the Sharpe ratio gain

$$\Delta\text{SR} = \text{SR}(r^A) - \text{SR}(r^B).$$

  The accompanying HAC $t$-statistic (Appendix I) applied to $\Delta_t$ tests whether the mean return differential differs from zero, rather than testing the equality of the two Sharpe ratios directly (Ledoit and Wolf, 2008).

- **Elementary score.** To measure forecast accuracy specifically at the decision threshold $\mathrm{IV}_t^2$, we evaluate the elementary score of Ehm et al. (2016) normalized by implied variance:

$$\mathrm{ES}_{1,t} = \left| \frac{\mathrm{RV}_t}{\mathrm{IV}_t^2} - 1 \right| \mathbf{1}\{\min(\hat{\sigma}_t^2, \mathrm{RV}_t) \le \mathrm{IV}_t^2 < \max(\hat{\sigma}_t^2, \mathrm{RV}_t)\},$$

  and denote its sample mean over $\mathcal{T}_{\mathrm{opt}}$ by $\overline{\mathrm{ES}}_1$.
- **Decision-threshold property.** Note that $\mathrm{ES}_{1,t}$ is strictly positive if and only if the forecast $\hat{\sigma}_t^2$ and the realization $\mathrm{RV}_t$ fall on opposite sides of the implied variance $\mathrm{IV}_t^2$—that is, precisely when the sign strategy takes the wrong directional position.
- **Compounding.** When evaluating compounded wealth trajectories from initial capital $V_0 = 1$, we allocate a fixed fraction $f = 3\%$ of current wealth to the straddle premium each day, yielding terminal portfolio value and annualized log growth rate

$$V_{T_{\mathrm{opt}}} = \prod_{t \in \mathcal{T}_{\mathrm{opt}}} (1 + f\, r_t), \qquad G = \frac{252}{T_{\mathrm{opt}}} \sum_{t \in \mathcal{T}_{\mathrm{opt}}} \log(1 + f\, r_t).$$

- Under bid/ask execution, $r_t^{\mathrm{ba}}$ replaces $r_t = q_t R_t$ in the compounding recursion. We define maximum drawdown as $\max_t(1 - V_t / \max_{t' \le t} V_{t'})$, where $V_t$ is the cumulative portfolio value at the end of day $t$.

### 7.3 Results

- **Unconditional short-straddle benchmark.** Selling the closing straddle unconditionally ($q_t \equiv -1$; labeled "Short straddle" in the tables) yields SR = 0.77 at mid-quotes and 0.42 at quoted bid/ask prices. Its mean daily return is statistically indistinguishable from zero (HAC $t = 0.43$), and the strategy loses money in 2 of the 4 calendar years: passively selling the closing 0DTE straddle earns no statistically detectable variance risk premium in our option-data subsample.
- **Sign strategies of the intraday models versus the short-straddle benchmark.** In point estimates, all 12 pooled intraday forecasts achieve a positive Sharpe ratio gain ($\Delta\mathrm{SR} > 0$) over the unconditional short straddle.

  **N15c.** Decision: to decide (text no longer matches the per-year Sharpe ratios of 2026-10-18).
  Problem: with the per-year annualization, 9 (not all) of the 12 pooled forecasts have $\Delta\mathrm{SR} > 0$ over the short straddle (this item); the full predictor set raises the pooled Sharpe ratio in 5 of the 6 classes, not all ($\Delta\mathrm{SR}$ from $-0.19$ for the random forest; the item below, 01_introduction, 09_conclusion "every intraday model class", and the abstract, paper_rehaul.tex, "across every intraday model class"); and the strongest pooled Sharpe ratios are elastic net, lasso, ridge, XGBoost, then LightGBM, so the list "elastic net, ridge, LightGBM" below no longer names the top three. The pooled XGBoost (1.26) is now above the pooled LightGBM (1.21), while the generator's cumulative-P&L and by-year figures (not in the paper) still show the LightGBM as the tree.

  - In 3 of the 12 specifications, the mean return gain over the short-straddle benchmark is statistically significant at the 5% level according to the HAC $t$-statistic, and all three use the full predictor set: LightGBM ($t = 2.05$), XGBoost ($t = 1.99$), and random forest ($t = 2.13$).
- **Highest Sharpe ratios among intraday models.** The strongest point-estimate Sharpe ratios among the pooled intraday specifications are all achieved on the full predictor set:
  - elastic net attains SR = 1.56 at mid-quotes and 1.23 at quoted bid/ask prices (with HAC $t = 1.90$ on its mean return differential against the short-straddle benchmark);

  - ridge attains 1.44 at mid-quotes and 1.10 at quoted bid/ask prices;
  - LightGBM attains 1.21 at mid-quotes and 0.87 at quoted bid/ask prices.
- **Full predictor set versus the HAR benchmark.** Expanding the information set from the HAR benchmark to the full predictor set increases the mid-quote Sharpe ratio in all 6 intraday model classes, shifting the range across classes from 0.74–1.22 to 1.00–1.56.
  - The resulting gain $\Delta$SR spans $-0.19$ to $+0.82$ across classes, with a median of $+0.38$ (elastic net $+0.82$, ridge $+0.69$, lasso $+0.72$).
  - Although economically sizable, the return differential is not statistically significant at the 5% level in any of the 6 classes (HAC $t$-statistic on the daily return differential).

**E16.** Decision: move "not significant" onto the mean return, or cite a Sharpe test.
Problem: this item says the Sharpe gain is not significant, and the HAC $t$ in parentheses tests the mean of the return gap $\Delta_t$. That is a different hypothesis.
This item says the full predictor set raises the Sharpe ratio of the sign strategy in all 6 classes, by $-0.19$ to $+0.82$, and that the gain is not significant in any class. The test in parentheses is a HAC $t$-statistic on the daily return gap $\Delta_t = r_t^A - r_t^B$, with Newey–West lag $q_{\mathrm{NW}} = 14$. Section 7.2 builds that $t$ as a test of the mean of $\Delta_t$, and says it does not test the equality of the two Sharpe ratios. The Sharpe gap $\mathrm{SR}(r^A) - \mathrm{SR}(r^B)$ divides each strategy's mean by its own standard deviation, so a test of the mean gap is a different statement from a test of the Sharpe gap. This item attaches "not significant" to the Sharpe gain.

- **Position sizing.** Across the 12 intraday forecasts, the median mid-quote Sharpe ratio is 1.20 under the sign strategy, 1.34 under the robust clip rule, and 1.40 under the percentile-rank rule.
  - In point estimates, continuous sizing via the robust clip outperforms the binary sign strategy ($\mathrm{SR}^{\mathrm{clip}} > \mathrm{SR}^{\mathrm{sign}}$) for 10 of the 12 intraday forecasts.
- **Concentration of economic gains.** As is typical of short-dated option strategies, cumulative performance is concentrated in a small subset of trading days:
  - removing each forecast's 10 best days renders the mean return differential $\bar{\Delta}$ over the short-straddle benchmark negative for 12 of the 12 intraday forecasts;
  - trading days on which the straddle expires worthless ($R_t = -1$, corresponding to settlement pinning near the strike; 137 of the 614 post-warmup days) account for 38% of the positive returns of the median pooled intraday forecast (34–44% across the 12 forecasts).
- **Statistical accuracy versus economic value.** Across the 12 pooled intraday forecasts evaluated on $\mathcal{T}_{\mathrm{opt}}$, the Spearman rank correlations between average statistical loss and the post-warmup Sharpe ratio (where a negative correlation indicates that lower loss corresponds to higher SR) are

$$\rho_S(\overline{\mathrm{QLIKE}}, \mathrm{SR}) = -0.52, \qquad \rho_S(\overline{\mathrm{ES}}_1, \mathrm{SR}) = -0.10.$$

  Among the pooled intraday forecasts, average QLIKE loss and $\overline{\mathrm{ES}}_1$, the loss at the implied-variance threshold, are both negatively rank-correlated with the Sharpe ratio, at $-0.52$ and a weak $-0.10$ for $\overline{\mathrm{ES}}_1$.
- **Robustness checks** (detailed in Appendix F):
  - *Interval-shift test:* when we replace $\hat{y}_{t,c}$ in the sign rule by the forecast $\hat{y}_{t,c+j}$ from $j$ half-hour intervals away ($j < 0$ earlier on day $t$; $j > 0$ on day $t+1$), $q_t = \mathrm{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \mathrm{IV}_t^2)$, the median Sharpe ratio across models varies between 0.00 and 1.58 across $j$ (a half-range of $\pm 0.79$). Because this variation lies well within the $\pm 1.06$ 95% sampling band of an annualized Sharpe ratio under i.i.d. returns, differences of this magnitude are consistent with sampling variation;

**E26.** Decision: to decide: restate on the pooled intraday forecasts only, or move to the closing-interval appendix.
Problem: the bullet reports a median Sharpe ratio across models without naming them, and the forecasts it shifts are not the pooled intraday forecasts of the tables.
The shift-test macros come from experiments/close_shift_test_headline.py through generated/close_shift_test_numbers.tex. Per results/close_shift_test/SUMMARY.md and shift_test_modelset.csv, the shifted forecasts are the "paper" family of the model set (a0, blk2, blk2_inc, lgbm, xgb, lasso_t, lasso_f, enet, b2lasso): the earlier draft's pooled 48-bar intraday forecasts. The current pooled linear rows (poolpt_*) and the per-bar rows are not shifted, because their tables hold only the 16:00 row (shift_test_not_shifted.csv). For $j > 0$ the shifted forecast is the one of day $t+1$, which is not in $\mathcal{F}_t$ at the 15:30 forecast origin, so those offsets are a placebo, and the bullet does not say so.

- *Subsample stability:* we verify performance across subsamples split around May 2022 (when CBOE expanded SPXW expirations to all five weekdays);
- *Compounded wealth:* across all 112 forecasts in the model set, compounding at $f = 3\%$ of wealth per day yields a median terminal portfolio value $V_{T_{\text{opt}}}$ of 4.4 at mid-quotes and 1.8 at quoted bid/ask prices, with maximum drawdowns reaching up to 64% and 78%, respectively.

**E27.** Decision: restrict the aggregation to intraday forecasts only (the 12 pooled intraday forecasts of note E25), dropping the closing-interval forecasts.
Problem: compounded wealth is computed across all 112 forecasts of the model set, closing-interval forecasts included, and the generator cannot yet compute it on the intraday forecasts of note E25.
The macros come from experiments/close_compounding_headline.py, which reads per-day returns from the master table (results/close_master_table/master_table_daily.parquet) for the forecasts of paper_model_set.csv. That set holds the per-bar trees and per-bar linear forecasts of the closing-interval scheme, the six pooled linear rows (poolpt_*), and the earlier draft's pooled 48-bar forecasts (the "paper" family). Of the six pooled trees, only the full-set LightGBM and XGBoost are in it, as the "paper" rows lgbm and xgb. The three HAR-benchmark pooled trees and the interim full-set random forest (results/pooled_trees_2026-10-07) are not in the master table, so a group or prefix filter cannot select the 12 forecasts. Missing: their per-day straddle returns in the compounding input, or a loader for those tables in the compounding script. The pooled-linear group alone is already in results/close_compounding/SUMMARY.md, with no macro.

Appendix G.5 reports each model class trained on the closing half hour alone.

# 8 Conclusion

- **Exogenous predictors reduce QLIKE loss in pooled intraday models.** Relative to the HAR benchmark, augmenting the model with the full exogenous predictor set lowers mean QLIKE across all 6 pooled intraday model classes, with 4 of these reductions statistically significant at the 5% level (Section 6). The largest relative reductions occur in the tree ensembles, while among the penalized linear specifications the elastic net achieves a statistically significant reduction.
- **Comparison with the closing-interval HAR benchmark.** Every pooled intraday specification incurs a higher mean QLIKE than the closing-interval HAR benchmark, the OLS HAR regression estimated solely on closing-interval data. Incorporating the full exogenous predictor set narrows this loss gap in every model class, leaving the gap statistically insignificant at the 5% level for all three penalized linear models while remaining significant for the tree ensembles (Section 6.4).
- **Models trained on the closing half hour.** Results for each model class estimated on the closing interval alone are reported in Appendix G.5.
- **Economic value in 0DTE straddles.** When the variance forecasts are used to trade same-day-expiring SPXW straddles over the closing half hour on the option-data subsample, adding the exogenous predictors increases the sign strategy's post-warmup Sharpe ratio in every intraday model class. Across the 12 pooled intraday forecasts, average QLIKE loss and $\overline{\mathrm{ES}}_1$, the loss at the implied-variance threshold, are both negatively rank-correlated with straddle Sharpe ratios, at $-0.52$ and a weak $-0.10$ for $\overline{\mathrm{ES}}_1$ (Section 7.3).

  **N15c.** Problem: "every intraday model class" no longer holds under the per-year Sharpe ratio (pooled random forest $\Delta\mathrm{SR} = -0.19$); see note N15c in Section 7.

- **Scope and limitations.**
  - Our out-of-sample evaluation spans $T_{\mathrm{oos}} = 1{,}082$ trading days from 3 January 2020 to 30 April 2024.
  - The empirical analysis focuses on a single equity index (the S&P 500) and a single target horizon (the 15:30–16:00 ET closing half hour).
  - In our economic evaluation, the straddle is held unhedged to cash settlement, and executing at quoted bid and ask prices rather than mid-quotes materially reduces the strategy's Sharpe ratio (Section 7).
- **Directions for future work.** Natural extensions include out-of-sample evaluation on post-30 April 2024 option expirations, cross-market validation on additional index option chains, and intraday variance forecasting at earlier decision origins prior to 15:30 ET.

# References


Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019.

Caio Almeida, Gustavo Freire, and Rodrigo Hizmeri. 0DTE asset pricing. `https://ssrn.com/abstract=4701401`, 2025. Working paper, SSRN 4701401.

Diego Amaya, Peter Christoffersen, Kris Jacobs, and Aurelio Vasquez. Does realized skewness

predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1):135–167, 2015. doi: 10.1016/j.jfineco.2015.02.009.

Torben G. Andersen and Tim Bollerslev. Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2-3):115–158, 1997.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003.

Torben G. Andersen, Tim Bollerslev, Peter F. Christoffersen, and Francis X. Diebold. Volatility and correlation forecasting. In *Handbook of Economic Forecasting*, volume 1, pages 777–878. Elsevier, 2006.

Torben G. Andersen, Tim Bollerslev, and Francis X. Diebold. Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4):701–720, 2007.

Federico M. Bandi, Nicola Fusari, and Roberto Renò. 0DTE option pricing. `https://ssrn.com/abstract=4503344`, 2023. Working paper, SSRN 4503344.

Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(2):253–280, 2002. doi: 10.1111/1467-9868.00336.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004.

Ole E. Barndorff-Nielsen, Silja Kinnebrock, and Neil Shephard. Measuring downside risk: Realized semivariance. In Tim Bollerslev, Jeffrey R. Russell, and Mark W. Watson, editors, *Volatility and Time Series Econometrics: Essays in Honor of Robert Engle*, pages 117–136. Oxford University Press, 2010. doi: 10.1093/acprof:oso/9780199549498.003.0007.

Heiner Beckmeyer, Nicole Branger, and Leander Gayda. Retail traders love 0DTE options... but should they? `https://ssrn.com/abstract=4404704`, 2023. Working paper, SSRN 4404704.

Jeremias Bekierman and Hans Manner. Forecasting realized variance measures using time-varying coefficient models. *International Journal of Forecasting*, 34(2):276–287, 2018.

Mikkel Bennedsen, Asger Lunde, and Mikko S. Pakkanen. Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006, 2022.

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

Tim Bollerslev, George Tauchen, and Hao Zhou. Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11):4463–4492, 2009. doi: 10.1093/rfs/hhp008.

Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016.

Tim Bollerslev, Benjamin Hood, John Huss, and Lasse Heje Pedersen. Risk everywhere: modeling and managing volatility. *The Review of Financial Studies*, 31(7):2729–2773, 2018.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

Jonathan Brogaard, Jaehee Han, and Peter Y. Won. Does 0DTE options trading increase volatility? `https://ssrn.com/abstract=4426358`, 2023. Working paper, SSRN 4426358.

Andrea Bucci. Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531, 2020. doi: 10.1093/jjfinec/nbaa008.

John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008. doi: 10.1093/rfs/hhm055.

Peter Carr and Liuren Wu. Variance risk premiums. *Review of Financial Studies*, 22(3):1311–1341, 2009. doi: 10.1093/rfs/hhn038.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Fulvio Corsi, Stefan Mittnik, Christian Pigorsch, and Uta Pigorsch. The volatility of realized volatility. *Econometric Reviews*, 27(1-3):46–78, 2008.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Chukwuma Dim, Bjorn Eraker, and Grigory Vilkov. 0DTEs: Trading, gamma risk and volatility propagation. `https://ssrn.com/abstract=4692190`, 2024. Working paper, SSRN 4692190.

Werner Ehm, Tilmann Gneiting, Alexander Jordan, and Fabian Krüger. Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(3):505–562, 2016.

Robert F. Engle and Giampiero M. Gallo. A multiple indicators model for volatility using intra-daily data. *Journal of Econometrics*, 131(1-2):3–27, 2006.

Robert F. Engle and Magdalena E. Sokalska. Forecasting intraday volatility in the US equity market. multiplicative component GARCH. *Journal of Financial Econometrics*, 10(1):54–83, 2012. doi: 10.1093/jjfinec/nbr005.

Jeff Fleming, Chris Kirby, and Barbara Ostdiek. The economic value of volatility timing. *Journal of Finance*, 56(1):329–352, 2001. doi: 10.1111/0022-1082.00327.

Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22, 2010. doi: 10.18637/jss.v033.i01.

Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.

Pierre Garrigues and Laurent El Ghaoui. An homotopy algorithm for the lasso with online observations. In *Advances in Neural Information Processing Systems*, volume 21, 2008.

Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Economic predictions with big data: The illusion of sparsity. *Econometrica*, 89(5):2409–2437, 2021.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Peter R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005. doi: 10.1198/073500105000000063.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79 (2):453–497, 2011.

Peter R. Hansen, Zhuo Huang, and Howard Howan Shek. Realized garch: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6):877–906, 2012.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Rodrigo Hizmeri and Mattia Bevilacqua. Early birds get the vol: Morning volatility uncertainty and variance risk premium. `https://ssrn.com/abstract=6212458`, 2025. Working paper, SSRN 6212458.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Serhiy Kozak, Stefan Nagel, and Shrihari Santosh. Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292, 2020.

Oliver Ledoit and Michael Wolf. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008. doi: 10.1016/j.jempfin.2008.03.002.

Robert C. Merton. Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987. doi: 10.2307/1913610.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.

Eghbal Rahimikia and Ser-Huang Poon. Machine learning for realised volatility forecasting. `https://ssrn.com/abstract=3707796`, 2020. Working paper, SSRN 3707796.

Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282, 2005. doi: 10.1111/j.1468-0262.2005.00615.x.

Zhouyu Shen and Dacheng Xiu. Can machines learn weak signals? Technical report, NBER Working Paper 33421, 2025.

Stephen J. Taylor and Xinzhong Xu. The incremental volatility information in one million foreign exchange quotations. *Journal of Empirical Finance*, 4(4):317–340, 1997. doi: 10.1016/S0927-5398(97)00010-8.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.

Mark A. van de Wiel, Mirrelijn M. van Nee, and Armin Rauschenberger. Fast cross-validation for multi-penalty high-dimensional ridge regression. *Journal of Computational and Graphical Statistics*, 30(4):835–847, 2021. doi: 10.1080/10618600.2021.1904962.

Wessel N. van Wieringen. Lecture notes on ridge regression, 2023. Version 0.60 (v8), 27 June 2023.

Grigory Vilkov. 0DTE trading rules: Tail risk, implementation, and tactical timing. `https://ssrn.com/abstract=4641356`, 2026. Working paper, SSRN 4641356; first posted 2023 as "0DTE Trading Rules".

Chao Zhang, Yihuang Zhang, Mihai Cucuringu, and Zhongmin Qian. Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22(2):492–530, 2024. doi: 10.1093/jjfinec/nbad005.

Hui Zou. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476):1418–1429, 2006. doi: 10.1198/016214506000000735.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B*, 67(2):301–320, 2005.

# A Appendix structure

- **Implementation details** (Appendix B): samples and windows, the training schemes in practice, the tuned hyperparameters and the tuning procedures.
- **Data and predictors** (Appendix C):
  - sources and spans;
  - predictor dictionary: 47 exogenous series and 634 intraday predictors in the full predictor set (47 series and 628 predictors in the closing-interval model) (Appendix Table 5);
  - missing and stale data;
  - removal of duplicate predictors; the window filter (constant and duplicate predictors dropped in each estimation window $\mathcal{W}_t$);
  - SPXW option chain; implied-volatility inversion;
  - FOMC calendar.
- **Hyperparameter selection by model** (Appendix D):
  - linear models: penalty strength $\lambda$, elastic-net mixing $\alpha$; block-penalized ridge, penalty weights $\omega_j$;
  - tree ensembles: fixed hyperparameters, Optuna budgets, selected trials; boosting rounds $b = 1, \ldots, B$, learning rate $\eta$;

  - re-selection frequency, on the estimation sample $\mathcal{W}_t^{\text{est}}$, gap $\mathcal{W}_t^{\text{gap}}$ and validation sample $\mathcal{W}_t^{\text{val}}$; re-estimation frequency;
- **Robustness, statistical** (Appendix E):
  - full model table, tree ensembles included (not reported);
  - Diebold–Mariano and HAC conventions (DM with the Harvey–Leybourne–Newbold correction; Newey–West lag truncation $q_{\text{NW}}$);
  - model confidence set, superior predictive ability, StepM;
  - regime and break tests (not reported);
  - closing-interval vs intraday models;
  - remaining variance through the trading day (not reported).
- **Robustness, trading** (Appendix F):
  - date-shift test;
  - Murphy diagrams (elementary scores); subsample splits;
  - P&L decomposition (not reported);
  - compounding;
  - position sizing;
  - strategy variations (not reported); other forecast origins; month-ends (not reported).
- **Closing-interval models** (Appendix G): the closing-interval sample (Appendix G.1), predictors and column counts (Appendix G.2), the closing-interval HAR benchmark and its estimation (Appendix G.3), closing-interval standardization (Appendix G.4), and every model class trained on the last 30 minutes (Appendix G.5).
- **Reproducibility** (Appendix H):
  - software lock files (not reported);
  - hashes of the design matrices (not reported);
  - computing environment;
  - CPU classes (not reported);
  - compute time (not reported);
  - the generator script behind every reported number.
- **Statistical tests** (Appendix I): the Diebold–Mariano statistic, the Newey–West variance and the HAC $t$-statistic.

**S13.** Decision: future work; keep the ten items in the overview, each marked (not reported), and report them properly in a later draft.
Problem: the overview lists ten items that the appendices do not report. Robustness, statistical (Appendix E): the full model table with the tree ensembles; regime and break tests; remaining variance through the trading day. Robustness, trading (Appendix F): the P&L decomposition; strategy variations; month-ends. Reproducibility (Appendix H): software lock files; hashes of the design matrices; CPU classes; compute time.

# B Implementation details

## B.1 Samples and windows

- **Estimation and evaluation samples:** Table 3.

**E31.** Decision: to decide: state the burn-in per panel in the samples note, as 3,125 30-minute grid intervals with its length in trading days, and fix it in make_rehaul_data_methods.py.
Problem: the samples note of Table 3 calls the burn-in 3,125 intervals for every panel, but on the closing-interval series, one row per trading day, it was dropped as 3,125 rows, that is 3,125 trading days, which is why that sample starts in 2010.
On the intraday grid 3,125 intervals are 3,125/48 trading days. The executor now drops the burn-in on the bar grid before the slice (src/backtest/executor.py:137–140, BURNIN_IN_BAR_SPACE, and 933–935), so a rerun of the closing-interval panel would start earlier than the printed sample. The note is generated by make_rehaul_data_methods.py, and re-running it rewrites generated/rehaul_dm_dictionary.tex, which carries the hand edit of note E5, so the note is not regenerated here. No macro holds the burn-in length in trading days.

**E20.** Decision: two window cells.
Problem: Panel B prints one intraday window, $W = 24{,}000$, and the ridge, lasso, and elastic-net QLIKE rows were fit on $W = 96{,}000$.
Panel B of Table 3 prints the intraday estimation sample as one row, and the window on that row is $W = 24{,}000$. Section B.2 and the note under the table give that window to the trees. Ridge, lasso, and elastic net are estimated on the longer window $W = 96{,}000$. The rolling-forecast row directly underneath is computed by subtracting 24,000 from the estimation-sample length, so the start date on that row is the first date at which a tree window is full. A linear window is full only after 96,000 intervals, and that origin is later. The QLIKE rows for ridge, lasso, and elastic net were computed from fits on the longer window. The cell they would be read against shows the shorter one.
Linked note (2026-10-17, user): E24 below, on the number in parentheses in Table 4, is the same underlying window confusion, a printed number read against the wrong estimation window (24,000 intervals, 96,000 intervals or 2,000 days).

- **Burn-in:** the first 3,125 samples (30-minute grid intervals), the longest HAR horizon, are dropped (Section 3).
- **Diurnal adjustment** $s^2_{t,i}$ (Section 5.1): $n_{\mathrm{d}} = 20$ trading days, at least $\underline{n}_{\mathrm{d}} = 5$ of them; intervals with fewer have $s^2_{t,i} = 1$ and are not scored.
- **Winsorization** of the target: $n_{\mathrm{w}} = 240$ intervals, all intervals of the day pooled; the bounds are the empirical 1st and 99th percentiles, $\underline{Q}_{t,i} = Q_{0.01}(\mathcal{Y}_{t,i})$ and $\overline{Q}_{t,i} = Q_{0.99}(\mathcal{Y}_{t,i})$.
- **Hyperparameter-selection splits:** $n_{\mathrm{gap}} = 25$ 30-minute intervals and $n_{\mathrm{val}} = 125$ 30-minute intervals.

**E21.** Decision: label each length with its unit and which step uses it.
Problem: $n_{\mathrm{gap}} = 25$ and $n_{\mathrm{val}} = 125$ are called observations, which on the intraday grid are 30-minute intervals, and the tuning text and Table 3 use the same symbols as numbers of trading days.
This item sets the embargo $n_{\mathrm{gap}}$ equal to 25 and the validation length $n_{\mathrm{val}}$ equal to 125, and it calls both of them observations. On the intraday grid an observation is one 30-minute interval, so an observation count is 48 times a count of trading days. The linear tuning item, a different step, describes its pre-sample in trading days: four folds of 250 days, running from 6 January 2016 to 2 January 2020. The light re-tune then says it uses two folds of 125 days, which puts the same symbol 125 on a count of days. Table 3 prints that pre-sample as four folds of 250 days after a gap of 25 days. Read as observations, 125 is a number of half hours. Read as in the re-tune and the table, it is a number of trading days.
Decision (2026-10-17, user): the main model is the intraday model, so both lengths are 30-minute intervals: $n_{\mathrm{gap}} = 25$ 30-minute intervals and $n_{\mathrm{val}} = 125$ 30-minute intervals, with no trading-day reference for either. Table 3 and its note, which are generated (`writeup/make_rehaul_data_methods.py`, near line 2283, "-day gap'), still print a 25-day gap.

## B.2 Training schemes in practice

- **Closing-interval scheme** ($\mathcal{I} = \{c\}$): Appendix G.3.
- **Intraday scheme** ($\mathcal{I} = \{1, \ldots, M\}$), the paper's training scheme: estimated on all $M = 48$ intervals of each day, overnight included; one observation per interval, so a period is an intraday interval and $\mathcal{W}_{t,i} = \{(t, i-W), \ldots, (t, i-1)\}$ holds the $W$ intervals before $(t, i)$. The full predictor set has 634 predictors (Section 4).

– Only the forecasts of the closing interval $i = c$, made at the 15:30 ET forecast origin, are scored and traded.
– Linear models: $W = 96{,}000$ intervals (2,000 trading days), re-estimated every interval.
– Trees: $W = 24{,}000$ intervals; one fixed hyperparameter configuration per tree method, never re-selected, re-estimated every interval.

### B.3 Tuned hyperparameters

- **Linear penalties,** chosen by hyperparameter selection on a pre-sample (Section B.4), one grid value per decade (Table 4):

$$\lambda^{\text{ridge}} \in \{10^{-2}, \ldots, 10^{3}\}, \qquad \lambda^{\text{lasso}}, \lambda^{\text{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  For lasso and elastic net these are the scikit-learn parameters (Section 5.2).
  – full predictor set: ridge $10^{3}$, lasso and elastic net $10^{-2}$ (top edge);
  – HAR benchmark: the lower edge of each grid.

  **N24.** Decision: freeze the pick (per E23): each penalty is the single pre-sample value in Panel A, held fixed with no re-tuning; the margin re-tune is not the rule.
  Problem: the current closing-interval linear forecasts still include the margin re-tune (4 of 24 re-selections replaced the penalty), and the Panel A table note describes it. A rerun with re-tuning switched off, using the Panel A picks, is pending and will produce the numbers to report. The table note comes from make_rehaul_data_methods.py, which cannot be re-run without wiping the E5 hand edit, so it must be parked or moved to another generator.

  **E23.** Decision: one pre-sample pick per estimator and predictor set, held fixed for the whole evaluation; no causal adaptive re-tuning during evaluation.
  Problem: the current results still include the margin re-tune: 4 of 24 re-selections replaced the HAR-benchmark penalty, so the closing linear forecasts must be re-run with re-tuning switched off for the text to match. The robustness claim below also needs a sensitivity check to back it.
  Each estimator and predictor set gets one penalty, chosen once by hyperparameter selection on the fixed pre-sample window and held constant at every forecast origin. Pre-tuning on a fixed window and holding the penalty constant is a simplification, and it is justified by robustness: the forecasts are not very sensitive to the penalty, so a penalty re-selected during the evaluation would change them little. The text does not claim that the exact value matters. The item above reports the pre-sample pick so that the forecasts can be reproduced, and the decision rests on that insensitivity, which a sensitivity check of the forecasts at neighboring grid values would have to confirm.

- **Trees,** fixed hyperparameters from a search on a 1998–2002 pre-sample (Table 4):
  – LightGBM and XGBoost: the best trial of an Optuna search (Akiba et al., 2019);
  – random forest: trial 255 of 400 ($B = 485$ trees, maximum depth 20, share of predictors tried at each split 0.90, bootstrap sample share 0.41);
  – minimum leaf size $n_{\min}$ (observations per leaf): the number outside the parentheses in Table 4 is the minimum leaf size used by the intraday trees on their 24,000-interval window (Section B.2); the number in parentheses is that minimum leaf size rescaled to the 2,000-day window of the closing-interval trees, $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, and it is used by the closing-interval trees only.

**E24.** Decision: the parenthetical is the closing-interval leaf size.
Problem: this sentence calls the rescaled $n_{\min}$ the intraday trees' window. The intraday fits use the unscaled $n_{\min}$, selected on 24,000 intervals. The number in parentheses is the closing-interval leaf size. The leaf minimum $n_{\min}$ was selected on a window of 24,000 intervals. This item rescales it by the ratio of window lengths, $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, and the sentence calls that product the window of the intraday trees. The note under Table 4 assigns the two numbers the other way around. The figure outside the parentheses is the minimum selected on the 24,000-interval window, and the figure inside the parentheses is that minimum rescaled to a window of 2,000 days. Section B.2 fits the intraday trees on 24,000 intervals, so those fits use the outside number, the one that was actually selected. The closing-interval trees are re-estimated every 10 days on a window of 2,000 days, and those are the fits that use the parenthetical.
Linked note (2026-10-17, user): E20 above, on the two window cells in Table 3, is the same underlying window confusion, a printed number read against the wrong estimation window (24,000 intervals, 96,000 intervals or 2,000 days).
Relabeled (2026-10-17, user): the leaf-minimum item above now calls the number in parentheses the closing-interval trees' minimum leaf size.

## B.4 Tuning procedures

- **Linear:**
  - hyperparameter selection on a pre-sample of four folds of 250 trading days (6 January 2016 to 2 January 2020), on validation MSE:

$$\hat{\lambda} = \arg\min_{\lambda} \frac{1}{4} \sum_{v=1}^{4} \mathrm{MSE}_v^{\mathrm{val}}(\lambda),$$

    with $\mathrm{MSE}_v^{\mathrm{val}}(\lambda)$ the mean squared error of $\tilde{y}$ on the validation sample of fold $v$;
  - a light re-tune at each re-tuning date (every 250 trading days): the grid neighbors $\mathcal{C} \subseteq \{\lambda^-, \lambda^+\}$ (one decade below and above, where on the grid) challenge the incumbent $\lambda$;
  - the best challenger $\lambda'$ (lowest validation MSE) replaces it only if the Diebold–Mariano statistic DM of Appendix I, on the paired squared errors, is significantly negative:

$$\mathrm{DM}(\{e_{t'}^2(\lambda') - e_{t'}^2(\lambda)\}_{t'}) < -z_{0.05/|\mathcal{C}|}, \qquad z_{0.05/|\mathcal{C}|} = \Phi^{-1}(1 - 0.05/|\mathcal{C}|);$$

  - $e_{t'}(\lambda) = \tilde{y}_{t'} - \hat{y}_{t'}(\lambda)$, $t'$ over two 125-interval (30-minute) validation folds; $\Phi$: the standard normal distribution function.
- **Trees,** in increasing order of tuning effort:
  - fixed hyperparameters, re-estimated every 10 trading days or daily;
  - random search every 250 trading days;
  - Optuna (Akiba et al., 2019): 50 trials, re-tuned every 1, 5, 25 or 250 trading days, best of the first 10, 25 or 50 trials.

**E22.** Decision: say which rung is in Table 1, and park the others.
Problem: the schemes section, the three-rung list in this subsection, and the draft or preliminary flag on the full-set tree rows each claim a tree cell in Table 1, and no cell is tied to one of them.
A tree cell in Table 1 is the loss of one fitting procedure, and three procedures are written down as if each of them were that procedure. Section B.2 gives each tree method a single configuration, chosen once and never re-selected, and re-estimates it every interval on a window of $W = 24{,}000$. The list in this subsection gives three grades of tuning instead: fixed hyperparameters re-estimated every 10 trading days or every day, a fresh random search every 250 trading days, and an Optuna search of 50 trials re-tuned every 1, 5, 25, or 250 trading days. Section 6.1 adds that the full-set LightGBM and XGBoost rows are an earlier draft and the random forest is preliminary, so those rows need not be any of the procedures just listed. Until one procedure is marked as the source of a given QLIKE cell, the cell has no window, no refit cadence, and no search attached to it.

- **Every selection uses only data available at the forecast origin;** no forecast is scored on a day that entered the selection of its own hyperparameters, and the evaluation span $\mathcal{T}_{\text{oos}}$ starts after the pre-sample.

# C Data and predictors

- **Sources and spans.** Eight groups, in Section 3. Closing-interval models use all 47 exogenous series. Intraday models use the 47 series in every group, FOMC included.
- **Predictor dictionary.** Appendix Table 5 defines each series. Trailing averages, activity indicators, and calendar columns are added in Section 4.
- **Missing and stale values.** For an exogenous series $u$, the availability indicator $a_{t,i} = \mathbf{1}\{u \text{ observed at } (t,i)\}$ is recorded before any filling (Section 3). The series is then forward-filled with no staleness limit. On every interval with $a_{t,i} = 0$ the stored regressor is the imputation of Section 4, step 4: the median of the series' observed adjusted values on earlier retained intervals, or the adjusted value on its first observed interval when no earlier observation exists.
- **Duplicate and constant predictors.** At each estimation the window screen of Section 4 drops a column that is constant on $\mathcal{W}_{t,i}$ or an exact bitwise copy of an earlier kept column. The first copy is kept. On the closing-interval model with the full predictor set the screen keeps a median of 369 of the 628 columns.
- **SPXW chain.** Quotes on S&P 500 Weeklys that expire the same day, every 30 minutes: bid, ask, mid-quote, underlying level, implied volatility, and delta. The file covers 1,291 expirations, 2020–2025. The straddle in Section 7.1 uses the 15:30 quotes. A day is dropped when the wider of the two strikes is more than 10 index points from $S_t$, or when fewer than 10 contracts are quoted at that stamp. $S_t$ is the 15:30 underlying print on the chain, and the official close is the S&P 500 close.
- **Implied-volatility inversion.** The vendor quote on each leg is a volatility per hour, from a geometric bisection. A solve that does not converge returns a bracket node of that bisection; those nodes are set to missing before the two legs are averaged, and a missing leg makes the package's implied variance missing. The 16:00 stamp is not used. The conversion from the hourly quote to the closing half hour is the display in Section 7.1.
- **FOMC calendar.** The four scheduled-announcement series of Section 3, on closing-interval models only. The schedule is known before the interval.

# D Hyperparameter selection by model

- **Penalty parameters.** Ridge, lasso, and elastic net minimize the raw sum of squares plus the penalty in Section 5.2. $\lambda$ is the coefficient on that penalty for ridge. For lasso and elastic net the printed grid is the scikit-learn parameter, and the coefficient in the sum is $2n$ times the printed value. Lasso is $\alpha = 1$. Elastic net is $\alpha = 0.5$. The block weights $\omega_j$ are 0.01 on the HAR benchmark, 1 on an additional predictor, and 0 on the intercept. The HAR benchmark fitted alone uses $\omega_j = 1$ on every predictor.
- **Where $\lambda$ is chosen.** One grid value per decade, Table 4. The pre-sample is four folds of 250 trading days, 6 January 2016 to 2 January 2020, and the criterion is validation MSE (Section B.4). The value is chosen once on this pre-sample and held fixed for the whole evaluation span, with no re-tuning during evaluation (note E23). The full predictor set stays

at the top of its grid. The HAR-benchmark column of the table is the pre-sample value. The current linear results still include an earlier margin re-tune, and its four replacements of the HAR-benchmark penalty are listed in note E23.

**E29.** Decision: state the yearly cadence and the closing-interval HAR-benchmark scope, consistent with E23's one pre-sample pick.
Problem: the light re-tune reads as frequent ("every 250 trading days" in the tuning procedures, Section B.4), but 250 trading days is a year: it ran on four dates, 23 June 2020 to 16 June 2023. It covers only the closing-interval linear arms (ridge, lasso and elastic net on the HAR benchmark and the full predictor set), and all 4 switches are on the HAR-benchmark set, in 2020 and 2023 (results/linear_pretune_2026-10-06/retune_log.csv, spec grid and margin rule; writeup/make_table_close_main.py, about lines 2303–2323). The tuning-procedure item states neither the yearly reading nor the scope.

- **Trees.** LightGBM and XGBoost take the best trial of an Optuna search on a 1998–2002 pre-sample. The random forest takes trial 255 of 400. The selected configuration and the leaf minimum are in Section B.3. Which of the three tuning rungs in Section B.4 produced a tree cell of Table 1 is note E22.
- **What may enter a choice.** A selection uses only data available at the forecast origin. The evaluation span starts after the pre-sample.

# E Robustness of the statistical results

- **Reported comparisons.** Forecast accuracy on $\mathcal{T}_{\text{oos}}$ is Table 1. The Diebold–Mariano statistic, with the Harvey–Leybourne–Newbold correction, and the Newey–West variance and HAC $t$-statistic are Appendix I.
- **Model confidence set, superior predictive ability, and StepM.** These three procedures are named in that appendix for the comparison of predictor sets. No table reports an MCS membership, an SPA $p$-value, or a StepM rejection (note E14).
- **Regime and break tests.** Regime tests and break tests have no reported output.
- **Closing-interval against intraday.** Both schemes forecast the same closing half hour. The comparison of their average QLIKE is in Section 6. A forecast of the variance remaining from an earlier origin to the close is not reported.

# F Robustness of the trade

- **Date-shift test.** The sign strategy is re-scored with the forecast of the interval $j$ half hours away from the closing interval, placed on the closing interval's diurnal adjustment:

$$q_t = \text{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \text{IV}^2_t).$$

$j < 0$ reads an earlier interval of day $t$. $j > 0$ reads day $t+1$. Across $j$, the median annualized Sharpe ratio runs from 0.00 to 1.58 (a swing of $\pm 0.79$), inside the $\pm 1.06$ 95% sampling band of an annualized Sharpe ratio under i.i.d. returns.

**N18.** Decision: to decide (data export needed).
Problem: the shift test runs on the earlier draft's nine pooled 48-bar forecasts (a0, blk2, blk2_inc, lgbm, xgb, lasso_t, lasso_f, enet, b2lasso); of the paper's 12 pooled intraday forecasts, only five have every intraday interval (the full-set LightGBM and XGBoost, which are the lgbm and xgb rows, and the three HAR-set pooled trees). The other seven, all six pooled linears (poolpt_*) and the interim full-set random forest, hold only the 16:00 row and cannot be shifted. Restating the test on the pooled forecasts needs an all-interval export of the six pooled linears (a cluster job; results/pool_linear_pretune_2026-10-06/forecasts.parquet has 16:00 only), the finished full-set random forest output, and a loader for results/pooled_trees_2026-10-07/ in experiments/close_shift_test_headline.py (its k = 0 checks also need a reference for those tables). See E26.

- **Murphy diagram.** The elementary score at the implied variance is $\mathrm{ES}_{1,t}$ in Section 7.2. Its rank correlation with the Sharpe ratio, across the 12 pooled intraday forecasts, is in Section 7.3. The subsample split is around May 2022.
- **Profit and loss.** The reported return is $R_t$ in Section 7.1. A decomposition of that return into separate variance and direction components is not reported.
- **Compounding.** Wealth starts at 1 and the premium each day is the fraction $f = 3\%$ in Section 7.2. Across the 112 forecasts, the median terminal value is 4.4 at mid-quotes and 1.8 at quoted bid and ask prices, and the maximum drawdowns reach 64% and 78%.
- **Position sizing.** The sign strategy and the two sized rules, the robust clip and the percentile rank, are in Section 7.1. All three are scored on the 614 days after the warmup of 252 days.
- **Other origins.** No other forecast origin is reported. The interval-shift test above keeps the 15:30 close, implied volatility and straddle, and takes only the model's forecast from another interval. A separate month-end strategy is not reported.

# G Closing-interval models

## G.1 Closing-interval sample

- The closing-interval specifications incorporate all 47 exogenous series, the FOMC group (4 series) included; so do the pooled intraday specifications, with 47 series (Section 3).
- The closing-interval sample ($i = c$) consists of 3,469 trading days from July 2010 through April 2024, estimated on a rolling window of $W = 2{,}000$ trading days.
- This rolling procedure yields 1,469 out-of-sample closing-interval forecasts starting in June 2018:

$$1{,}469 = 3{,}469 - 2{,}000.$$

- Hyperparameter selection for the linear specifications is conducted on an initial pre-sample using closing-interval observations starting on 13 April 2004, with validation windows spanning 6 January 2016 to 2 January 2020.

## G.2 Closing-interval predictors

- The horizon set of the closing-interval model is base 5,

$$\mathcal{K} = \{1, 5, 25, 125, 625, 3125\},$$

  the same set as the intraday model. Here, $k$ counts retained intervals across the full intraday grid $\mathcal{G}$ (including overnight sessions), while the closing-interval specification restricts estimation to the rows $i = c$.
- In the closing-interval panel, the $k$-interval trailing averages of the activity indicator, $\bar{\chi}^{(k)}_{t,i}$, are computed for 7 mode-inflated series (note E6).

- Under the closing-interval specification with the full predictor set, the window filter retains a median of 369 out of 628 candidate columns across rolling windows.
- **Predictor set dimensions.** In the closing-interval specification, the HAR benchmark contains 22 candidate columns, whereas the full predictor set contains 628 columns:

$$628 = \underbrace{22}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{47 \times 6}_{\bar{u}^{(k)}} + \underbrace{47 \times 6}_{\bar{a}^{(k)}} + \underbrace{7 \times 6}_{\bar{\chi}^{(k)}}, \qquad 22 = 6 + 16.$$

## G.3 Closing-interval HAR benchmark and estimation

- **Closing-interval HAR benchmark.** The benchmark forecast of the main text is the daily HAR regression of Corsi (2009), estimated by OLS on the HAR components and calendar dummies (Section 5.2) using closing-interval observations only: one observation per trading day, re-estimated daily on a rolling window of $W = 2{,}000$ trading days.
- Under the closing-interval scheme, a period is a trading day and the target is $y_t \equiv y_{t,c}$; the refit cadence of each model class is in the next item.
- **Closing-interval scheme:** $W = 2{,}000$ trading days for every model class; re-estimated every trading day, except the tree specifications re-estimated every 10 trading days (fixed hyperparameters or random search).

**E32.** Decision: user decision 2026-10-18: the tree specifications refit every session (every trading session), not every 10 trading days; "daily" applies only to the closing-interval linear models. The rerun is still pending.
Problem: the closing-interval tree specifications refit every 10 trading days (specs/causal_tune_trees.py:35–40, REFIT_EVERY = 10 sessions; jobs/sge/unification_tree.sge:17, TREE_REFIT_EVERY=10), so their printed numbers are not every-session refits.
Requires a rerun of the closing-interval tree specifications that refit every 10 trading days, with an every-session refit (the fixed-hyperparameter rung, subtree_daily_*, already refits every session: experiments/close_studies_export.py:75 and 80, refit_every = 1). The pooled intraday models already refit at every interval (experiments/pool_linear_pretune.py:8–9 and specs/causal_tune_linear.py:159, REFIT_FREQUENCY = 1; experiments/pooled_trees_score.py:139–140), and the closing-interval linear models and the HAR benchmark every trading day (experiments/close_linear_pretune.py:11).

## G.4 Closing-interval standardization

The closing-interval model of Section G.5 standardizes the averages of series values on its own window $\mathcal{W}_t$ (Section B.2). Rows $t'$ are trading days ($i = c$ suppressed):

$$z_t^{(k)} = \frac{\bar{u}_t^{(k)} - \mathrm{med}_{t' \in \mathcal{W}_t}\, \bar{u}_{t'}^{(k)}}{\max\{\mathrm{IQR}_t^{(k)},\ 0.01 \max_{t' \le t} \mathrm{IQR}_{t'}^{(k)},\ \mathrm{IQR}_t^{*(k)}\}}, \qquad \mathrm{IQR}_t^{(k)} = \mathrm{IQR}\{\bar{u}_{t'}^{(k)} : t' \in \mathcal{W}_t\}.$$

- $\mathrm{IQR}_t^{*(k)}$: IQR of $\bar{u}_{t'}^{(k)}$, $t' < t$, over rows whose $k$-interval averaging window is fully observed.
- Column whose mode accounts for at least 20% of rows: $\mathrm{IQR}_t^{*(k)}$ also excludes rows at the mode.
- $\mathrm{IQR}_t^{*(k)} = 1$ until 8 such rows exist.
- $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized. HAR components, calendar dummies: $\mathrm{IQR}_t^{*(k)} = 0$.
- $t < W$: the first window's statistics.

## G.5 Training on the last 30 minutes

- **Closing-interval estimation scheme ($\mathcal{I} = \{c\}$).** As a final refinement, we restrict estimation of each model class in Section 5.2 exclusively to observations from the closing half hour $i = c$ (15:30–16:00 ET), matching the estimation domain of the daily HAR benchmark.
  - Under this scheme, each trading day contributes a single observation, so a period corresponds to one trading day and the rolling window $\mathcal{W}_t = \{t - W, \ldots, t - 1\}$ spans $W = 2{,}000$ trading days (Appendix B.2). Rolling standardization of the exogenous series averages on $\mathcal{W}_t$ follows Appendix G.4.
  - We hold the predictor sets, evaluation sample $\mathcal{T}_{\text{oos}}$, and straddle trading rules unchanged; statistical accuracy is reported in Table 2 over the same forecast origins as Table 1.
- **Improvements in QLIKE accuracy.** Training solely on the closing interval reduces mean QLIKE relative to pooled intraday training in every model class and under both predictor sets. Specifically, the percentage excess loss of the pooled intraday forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$ relative to the closing-interval forecast $f_{m,\mathcal{X},\{c\}}$, $\Delta\% = 100(\bar{L}^{\text{pool}}/\bar{L}^{\text{close}} - 1)$, is strictly positive throughout:
  - for penalized linear models, pooling increases mean QLIKE by +10.0% to +10.7% on the HAR benchmark (DM from 3.09 to 3.21, all significant at the 5% level) and by +5.3% to +11.1% on the full predictor set (significant for the lasso);
  - for tree ensembles, pooling increases mean QLIKE by +18.5% to +21.7% on the HAR benchmark and by +7.5% to +9.1% on the full predictor set, with every difference significant at the 5% level.
- **Preservation of the exogenous-predictor gains.** Within the closing-interval scheme, expanding from the HAR benchmark to the full predictor set lowers mean QLIKE in all 6 model classes, with relative reductions $\Delta\%$ between $-4.1\%$ and $-0.1\%$, although none of the single-class reductions is individually significant at the 5% level (for lasso, $\Delta\% = -4.0\%$, $\text{DM} = -1.91$, $p = 0.056$).
  - Across all 51 matched predictor-set pairs in the full model set (45 tree and 3 linear specifications trained on the closing interval, plus 3 pooled intraday specifications), adding the exogenous predictors reduces mean QLIKE in 35 pairs, 4 of which are significant at the 5% level.
- Across the 12 pooled intraday forecasts, 0 achieve a lower mean QLIKE than the daily HAR benchmark (0 significantly), while 9 perform significantly worse; the closing-interval forecasts are left out of this count.
- Comparing tree ensembles directly against the closing-interval ridge regression on the same predictor set, the tree specification incurs a higher mean QLIKE in 48 of 48 pairs (23 significantly).
- **Sensitivity of squared-error loss to extreme days.** In contrast to QLIKE, evaluated under MSE the full predictor set increases squared-error loss in 49 of the 51 matched pairs across the model set (0 significantly).
  - A single date outside the option-data subsample, 19 March 2020, accounts for 28–39% of the total cumulative squared error of the closing-interval tree models over $\mathcal{T}_{\text{oos}}$, compared with at most 0.9% of their total QLIKE loss.
  - Similarly, five trading days in March 2020 account for 73–87% of the cumulative squared error of the daily HAR benchmark and of the closing-interval lasso and XGBoost on the full predictor set; when February–April 2020 are excluded, the closing-interval lasso

achieves a lower MSE than the benchmark forecast.

- **Economic value under the closing-interval scheme** (sign strategy evaluated on the 614 post-warmup days of $\mathcal{T}_{\text{opt}}$):
  - The closing-interval elastic net on the full predictor set attains SR = 1.76 at mid-quotes and 1.42 at quoted bid/ask prices, a statistically significant mean return gain over the short-straddle benchmark (HAC $t = 2.56$). The closing-interval lasso on the full predictor set attains SR = 1.68 at mid-quotes and 1.34 at quoted bid/ask prices.
  - Closing-interval LightGBM on the full predictor set, with fixed hyperparameters re-estimated every trading day (the specification of the closing-interval tree rows of the main results table), attains SR = 1.38 at mid-quotes; the tree rows of the straddle table are the pooled intraday trees.
  - Replacing the HAR benchmark with the full predictor set changes the closing-interval sign-strategy Sharpe ratio by $\Delta$SR = +0.19 for the elastic net and +0.12 for the lasso (neither statistically significant at the 5% level); across all 51 matched predictor-set pairs in the model set, $\Delta$SR $> 0$ in 49 pairs (median +0.43), with 4 significant at the 5% level.
  - Relative to the unconditional short-straddle benchmark, $\Delta$SR $> 0$ for 78 of the 112 forecasts in the model set, and all 20 gains that are statistically significant at the 5% level use the full predictor set.
  - Finally, none of the return differentials between a pooled intraday model and the corresponding closing-interval model of the same class is statistically significant at the 5% level (HAC $t$-statistic on the daily return differential with $q_{\text{NW}} = 14$).

**Table 2:** Forecast accuracy when training on the last 30 minutes

| | HAR benchmark | | | All features | | |
|---|---|---|---|---|---|---|
| Model | MSE ($10^{-10}$) | QLIKE | $R^2_{\text{OOS}}$ (%) | MSE ($10^{-10}$) | QLIKE | $R^2_{\text{OOS}}$ (%) |
| OLS (benchmark forecast) | 4.15 | 0.1033 | 78.2 | – | – | – |
| *Panel A: Linear models* | | | | | | |
| Ridge | 4.15 | 0.1033 | 78.2 | 7.30 | 0.1031 | 61.7 |
| Lasso | 4.15 | 0.1033 | 78.2 | 5.94 | 0.0992 | 68.8 |
| Elastic net | 4.15 | 0.1032 | 78.2 | 6.89 | 0.1014 | 63.9 |
| *Panel B: Tree ensembles* | | | | | | |
| LightGBM | 5.75 | 0.1094 | 69.8 | 17.19 | 0.1049 | 9.8 |
| XGBoost | 5.27 | 0.1076 | 72.4 | 20.94 | 0.1031 | −9.9 |
| Random forest | 4.62 | 0.1068 | 75.8 | 14.15 | 0.1060 | 25.8 |

*Notes:* Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, $\text{RV}_t$, on the $T_{\text{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 5.3). The span starts with the intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). The MSE entries are in units of $10^{-10}$: the printed number times $10^{-10}$ is the mean squared error $T_{\text{oos}}^{-1}\sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2$. QLIKE is the mean QLIKE loss. $R^2_{\text{OOS}} = 1 - \sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\text{RV}_t - \overline{\text{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\text{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 4; the OLS is estimated on the HAR benchmark only (–: no forecast). Each model is estimated on the last 30 minutes of each trading day only (one observation per day); the OLS on the HAR benchmark is the benchmark forecast. Trees use fixed hyperparameters and are re-estimated daily (Section B.2). The intraday models on the same days are in Table 1.

# H Reproducibility

- **Generator of each reported number.**
  - `writeup/make_rehaul_data_methods.py`: samples, windows, hyperparameter tables, and the predictor dictionary.
  - `writeup/make_rehaul_oos_table.py`: Table 1 and the last-30-minutes split.
  - `writeup/make_rehaul_results.py`: the accuracy macros in Section 6.
  - `writeup/make_rehaul_economics.py`: the straddle macros in Section 7.
  - `writeup/make_rehaul_fi.py`: the penalty weight on the HAR block in Appendix D and the closing-interval forecast count in Appendix G.1.
  - `writeup/make_rehaul_appendix.py`: the appendix-table macros.
  - `writeup/make_table_close_main.py`: the closing-interval master-table macros, including the closing-interval predictor counts.
  - `writeup/make_fig_qlike_sharpe.py`: the matched predictor-set pair counts and the Newey–West lag of the HAC $t$-statistics.
  - `writeup/make_buckets_tex.py`: the group sizes and the exogenous series count.
  - `experiments/close_murphy_splits_headline.py`: the Murphy-diagram macros.
  - `experiments/close_compounding_headline.py`: the compounding macros in Appendix F.
  - `experiments/close_shift_test_headline.py`: the interval-shift test macros in Appendix F.
- **Software list.** `requirements.txt` at the repository root names the packages and a version range for each. It is not a pinned lock file, and the draft does not record a second lock file.
- **Design matrices, CPU class, and compute time.** The generators read the forecasts and tables already on disk. The draft does not record a hash of the design matrix behind Table 1, the CPU class of those fits, or their compute time.

**Table 3:** Estimation and evaluation samples.

| Sample | Start | End | Obs. | Window $W$ |
|---|---|---|---|---|
| Intraday grid: 30-minute intervals with an observed $RV_{t,i}$, outside the weekend break | 1998-01-05 | 2024-04-30 | 303,442 | – |
| *Panel A: Closing-interval models (trading days)* | | | | |
| Closing-interval observations, after the burn-in | 2010-07-12 | 2024-04-30 | 3,469 | 2,000 |
| Rolling-window forecasts | 2018-06-25 | 2024-04-30 | 1,469 | – |
| Hyperparameter-selection pre-sample (linear models) | 2004-04-13 | 2020-01-02 | 3,947 | 2,000 |
| Validation samples ($4 \times 250$ days) | 2016-01-06 | 2020-01-02 | 1,000 | 2,000 |
| *Panel B: Intraday models (30-minute intervals)* | | | | |
| Estimation sample, after the 3,125-interval burn-in | 1998-06-04 | 2024-04-30 | 300,317 | 24,000 |
| Rolling-window forecasts | 2001-07-13 | 2024-04-30 | 276,317 | – |
| Tree hyperparameter pre-sample (before 2003) | 1998-06-04 | 2002-12-31 | 36,285 | 24,000 |
| *Panel C: Evaluation samples (trading days)* | | | | |
| Evaluation span ($T_{\text{oos}}$): trading days on which every forecast exists, less early-close days (6) | 2020-01-03 | 2024-04-30 | 1,082 | – |
| Option-data subsample ($T_{\text{opt}}$): SPXW expiration days (872) less early-close days (6) | 2020-01-03 | 2024-04-30 | 866 | – |
| Later SPXW expiration days, held out (413 full trading days) | 2024-05-01 | 2025-12-31 | 419 | – |

*Notes:* Dates are interval-end dates (ET); Obs. and the rolling estimation window $W$ are in the units of each panel. The burn-in, 3,125 intervals, is the longest HAR horizon and is applied to each series as built (the closing-interval series, the intraday grid and, for the linear pre-sample, the series of the two intervals of the last hour), so the closing-interval sample starts in 2010 and the pre-sample in 2004. The pooled linear models use $W = 96{,}000$ intervals (2,000 trading days); the tree hyperparameters are chosen once, by an Optuna search on the 1998–2002 development sample that fits each candidate on the previous 24,000 intervals and scores it on the next 250, stepping forward 250 intervals at a time; the reported forecasts refit every interval. Each validation sample follows a 25-day gap and a 2,000-day estimation sample (Section B.1). Full trading days: expiration days that are not early-close days. Panel C: the evaluation span holds every trading day on which all rolling forecasts exist (Section 5.3); the straddle economics use its option-data subsample (Section 7).

**Table 4:** Hyperparameters: penalized regressions and tree ensembles.

| | | | Value | |
|---|---|---|---|---|
| Model | Hyperparameter | Search space | HAR benchmark | Full predictor set |
| *Panel A: Penalized linear models, selected on the pre-sample* | | | | |
| OLS | – | – | – | – |
| Ridge | Penalty, $\lambda$ | $\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6) | $10^{-2}$ | $10^{3}$ |
| Lasso | Penalty, $\lambda$ | $\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5) | $10^{-6}$ | $10^{-2}$ |
| Elastic net | Penalty, $\lambda$ | $\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5) | $10^{-6}$ | $10^{-2}$ |
| | Mixing weight, $\alpha$ | fixed | 0.5 | 0.5 |
| All three | Penalty weight on the HAR benchmark, $\underline{\omega}$ | fixed | 1 | 0.01 |
| *Panel B: Tree ensembles, fixed hyperparameters (selected on the 1998–2002 pre-sample)* | | | | |
| LightGBM | Leaves per tree, $J$ | [15, 255] (log) | 20 | |
| | Learning rate, $\eta$ | [0.01, 0.3] (log) | 0.013 | |
| | Boosting rounds, $B$ | [50, 500] | 392 | |
| | Minimum leaf size, $n_{\min}$ | [20, 500] (log) | 98 (8) | |
| | Share of predictors per tree | [0.5, 1] | 0.70 | |
| | Share of observations per tree | [0.5, 1] | 0.56 | |
| | $L_1$ penalty on leaf weights, $\lambda_1^{\text{tree}}$ | $[10^{-8}, 10]$ (log) | $10^{-8}$ | |
| | $L_2$ penalty on leaf weights, $\lambda_2^{\text{tree}}$ | $[10^{-8}, 10]$ (log) | 0.016 | |
| XGBoost | Maximum depth | [3, 10] | 5 | |
| | Learning rate, $\eta$ | [0.01, 0.3] (log) | 0.0123 | |
| | Boosting rounds, $B$ | [50, 500] | 367 | |
| | Minimum leaf size, $n_{\min}$ | [1, 100] (log) | 50.2 (4.18) | |
| | Share of observations per tree | [0.5, 1] | 0.66 | |
| | Share of predictors per tree | [0.5, 1] | 0.65 | |
| | $L_1$ penalty on leaf weights, $\lambda_1^{\text{tree}}$ | $[10^{-8}, 10]$ (log) | 0.0215 | |
| | $L_2$ penalty on leaf weights, $\lambda_2^{\text{tree}}$ | $[10^{-8}, 10]$ (log) | $1.2 \times 10^{-5}$ | |
| | Penalty per leaf, $\gamma$ | $[10^{-8}, 1]$ (log) | $1.5 \times 10^{-6}$ | |
| Random forest | Number of trees, $N$ | [50, 500] | 485 | |
| | Maximum depth | [2, 32] (log) | 20 | |
| | Share of predictors per split | [0.02, 1] (log) | 0.90 | |
| | Bootstrap sample share | [0.05, 1] (log) | 0.41 | |
| | Minimum leaf size, $n_{\min}$ | [1, 1000] (log) | 12 (1) | |

*Notes:* Panel A: Section 5.2 states the scale of $\lambda$. One grid value per decade (grid size in parentheses), selected by mean validation MSE over the 4 folds of the pre-sample (Table 3). At each of the 4 re-selection points a grid neighbor may replace it (Section B.4); this happened 4 times in 24 re-selections (ridge 1 and 0, lasso 1 and 0, elastic net 2 and 0, for the HAR benchmark and the full predictor set). Panel B: the best trial of an Optuna search on the intraday pre-sample (24,000-interval window; random forest: trial 255 of 400), ranked by MSE and then held fixed for both predictor sets; (log): searched on a logarithmic scale. $n_{\min}$ was selected on 24,000 intervals and is rescaled to the 2,000-day window (rescaled value in parentheses). Implementations: LightGBM (Ke et al., 2017) with bagging at every round, XGBoost (Chen and Guestrin, 2016) with histogram-based split finding and the scikit-learn random forest with bootstrap samples; random seeds are fixed.

**Table 5:** Predictor definitions.

| Variable | Description | Source | Diurnal adjustment | Transform $g(v)$ | Predictors, $k \in \mathcal{K}$ |
|---|---|---|---|---|---|
| *Realized moments (7; front-month E-mini, one-minute log-price changes)* | | | | | |
| `sumret` | Sum of one-minute changes in the log price of the front-month contract, $\texttt{sumret}_{t,i} = \sum_k r_{t,i,k}$ | E-mini S&P 500 futures | s.d. | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumabsret` | Realized absolute variation, $\sum_k \lvert r_{t,i,k}\rvert$ | E-mini S&P 500 futures | mean | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret3` | Realized skewness, unnormalized, $\sum_k r_{t,i,k}^3$ | E-mini S&P 500 futures | s.d. | $v^{1/3}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret4` | Realized quarticity, unscaled, $\sum_k r_{t,i,k}^4$ | E-mini S&P 500 futures | mean | $\text{sign}(v)\,\lvert v\rvert^{1/4}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumpret2` | Positive realized semivariance, $\sum_k r_{t,i,k}^2\,\mathbf{1}\{r_{t,i,k} > 0\}$ | E-mini S&P 500 futures | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumbipow` | Bipower variation, $\texttt{sumbipow}_{t,i} = \sum_{k\geq 2} \lvert r_{t,i,k}\rvert\,\lvert r_{t,i,k-1}\rvert$ | E-mini S&P 500 futures | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumautocov` | First-order realized autocovariance, $\texttt{sumautocov}_{t,i} = \sum_{k\geq 2} r_{t,i,k}\,r_{t,i,k-1}$ | E-mini S&P 500 futures | s.d. | $\text{sign}(v)\,\lvert v\rvert^{1/2}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Liquidity (14)* | | | | | |
| `sumvolume` | Number of front-month contracts traded in the half hour | E-mini S&P 500 futures | mean | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `numobs` | Number of non-missing one-minute intervals in the half hour, $\texttt{numobs}_{t,i} \leq 30$; mode 30 on 72% of intervals | E-mini S&P 500 futures | mean | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| `turnover_ewstock` | Equal-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `buyturnover_ewstock` | Equal-weighted buyer-initiated turnover, the same ratio using estimated buy orders | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sellturnover_ewstock` | Equal-weighted seller-initiated turnover, the same ratio using estimated sell orders | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `effspread_ewstock` | Equal-weighted effective spread, the mean percent difference between the trade price and the preceding midquote | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `spread_ewstock` | Equal-weighted quoted spread, the mean percent difference between the bid and the ask | Constituents | mean | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `turnover_vwstock` | Value-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |

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*Table 5 (continued)*

| Variable | Description | Source | Diurnal adjustment | Transform $g(v)$ | Predictors, $k \in \mathcal{K}$ |
|---|---|---|---|---|---|
| `buyturnover_vwstock` | Value-weighted buyer-initiated turnover, the same ratio using estimated buy orders | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sellturnover_vwstock` | Value-weighted seller-initiated turnover, the same ratio using estimated sell orders | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `effspread_vwstock` | Value-weighted effective spread, the mean percent difference between the trade price and the preceding midquote | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `spread_vwstock` | Value-weighted quoted spread, the mean percent difference between the bid and the ask | Constituents | s.d. | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `ofi_ewstock` | Equal-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover | Constituents | s.d. | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `ofi_vwstock` | Value-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover | Constituents | s.d. | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Equal-weighted market (6; simple averages of S&P 500 constituents)* | | | | | |
| `sumret2_ewstock` | Equal-weighted realized variance of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret3_ewstock` | Equal-weighted realized skewness of 5-minute log returns, unnormalized | Constituents | s.d. | $v^{1/3}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret4_ewstock` | Equal-weighted realized quarticity of 5-minute log returns, unscaled | Constituents | mean | $\text{sign}(v)\,\lvert v\rvert^{1/4}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumabsret_ewstock` | Equal-weighted realized absolute variation of 5-minute log returns | Constituents | mean | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumbipow_ewstock` | Equal-weighted bipower variation of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumpret2_ewstock` | Equal-weighted positive realized semivariance of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Value-weighted market (6; previous day's market capitalization)* | | | | | |
| `sumret2_vwstock` | Value-weighted realized variance of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret3_vwstock` | Value-weighted realized skewness of 5-minute log returns, unnormalized | Constituents | s.d. | $v^{1/3}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumret4_vwstock` | Value-weighted realized quarticity of 5-minute log returns, unscaled | Constituents | mean | $\text{sign}(v)\,\lvert v\rvert^{1/4}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |

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*Table 5 (continued)*

| Variable | Description | Source | Diurnal adjustment | Transform $g(v)$ | Predictors, $k \in \mathcal{K}$ |
|---|---|---|---|---|---|
| `sumabsret_vwstock` | Value-weighted realized absolute variation of 5-minute log returns | Constituents | mean | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumbipow_vwstock` | Value-weighted bipower variation of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `sumpret2_vwstock` | Value-weighted positive realized semivariance of 5-minute log returns | Constituents | mean | $\sqrt{v}$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Sentiment (3)* | | | | | |
| `stocktwits_attention` | Number of StockTwits posts in the half hour | StockTwits | mean | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `stocktwits_sentiment` | Score of tagged posts, 1 if all are bullish and $-1$ if all are bearish | StockTwits | none | $v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `stocktwits_sentcount` | Number of posts in the half hour that carry a bullish or bearish tag | StockTwits | mean | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Implied volatility (3)* | | | | | |
| `vix` | VIX index (30-day S&P 500 implied volatility) | Cboe | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `vvix` | VVIX index (implied volatility of VIX) | Cboe | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `vix3m` | VIX3M index (3-month S&P 500 implied volatility) | Cboe | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| *Volatility demand (4; net vega bought by non-market-makers)* | | | | | |
| `voldemand_spx_open_and_close` | Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open or close a position; mode 0 on 59% of intervals | Cboe | none | $\operatorname{asinh}(u/\kappa)$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| `voldemand_spx_open_only` | Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open a position; mode 0 on 60% of intervals | Cboe | none | $\operatorname{asinh}(u/\kappa)$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| `voldemand_all_open_and_close` | Net vega bought by non-market-makers, all traded options, trades that open or close a position; mode 0 on 59% of intervals | Cboe | none | $\operatorname{asinh}(u/\kappa)$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| `voldemand_all_open_only` | Net vega bought by non-market-makers, all traded options, trades that open a position; mode 0 on 60% of intervals | Cboe | none | $\operatorname{asinh}(u/\kappa)$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| *FOMC (4; scheduled calendar)* | | | | | |
| `fomc_release` | FOMC announcement-interval dummy (14:30 ET); mode 0 on 98% of intervals | FOMC calendar | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |

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*Table 5 (continued)*

| Variable | Description | Source | Diurnal adjustment | Transform $g(v)$ | Predictors, $k \in \mathcal{K}$ |
|---|---|---|---|---|---|
| `fomc_day` | FOMC announcement-day dummy; mode 0 on 95% of intervals | FOMC calendar | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$ |
| `fomc_until_inv` | $1/(1+n^{+}_{t,i})$, $n^{+}_{t,i}$ intervals to the next announcement | FOMC calendar | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |
| `fomc_since_inv` | $1/(1+n^{-}_{t,i})$, $n^{-}_{t,i}$ intervals since the last announcement | FOMC calendar | none | $\log v$ | $u^{(k)}$, $\bar{a}^{(k)}$ |

*Notes:* Each series $u$ enters as $u^{(k)}$, its adjusted, transformed and winsorized value averaged over the previous $k$ intervals, and as $\bar{a}^{(k)}(u)$, the share of those intervals on which it was observed; series with an activity indicator also enter as $\bar{\chi}^{(k)}(u)$, the share off their mode (the mode and its share of intervals are in the description). $\mathcal{K} = \{1, 5, 25, 125, 625, 3125\}$ in the closing-interval model and in the intraday model; both models use every group here (Section 3). Diurnal adjustment: division by the trailing same-interval mean (mean) or standard deviation (s.d.); $g$: the variance-stabilizing map; the volatility-demand series take $\operatorname{asinh}(u_{t,i}/\kappa_{t,i})$ in place of both (Section 4). A series is signed if it has any negative value: `spread_vwstock` has a single negative value (2018-11-12 11:00 ET) and so takes $g(v) = v$ and the s.d. adjustment, where `spread_ewstock` takes $\log v$ and the mean. Unnormalized and unscaled: without the factors $\sqrt{N_{t,i}}/\mathrm{RV}_{t,i}^{3/2}$ (skewness), $N_{t,i}/3$ (quarticity) and $\pi/2$ (bipower variation).

# I Statistical tests

- **Diebold–Mariano test.** To compare the predictive accuracy of two competing forecasts $A$ and $B$, we apply the Diebold and Mariano (1995) test to the daily QLIKE loss differentials, incorporating the small-sample adjustment of Harvey et al. (1997) for one-step-ahead forecasts:

$$d_t = L_t^A - L_t^B, \qquad \mathrm{DM} = \sqrt{\frac{|\mathcal{O}|-1}{|\mathcal{O}|}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/|\mathcal{O}|}}.$$

  - Here $A$ and $B$ denote competing forecast series $f_{m,\mathcal{X},\mathcal{I}}$, indexed by the model class $m$, the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$, and the estimation interval set $\mathcal{I} \in \{\{c\}, \{1,\dots,M\}\}$.
  - The evaluation set $\mathcal{O}$, indexed chronologically by $t = 1,\dots,|\mathcal{O}|$, equals the full evaluation span $\mathcal{O} = \mathcal{T}_{\mathrm{oos}}$ ($|\mathcal{T}_{\mathrm{oos}}| = T_{\mathrm{oos}}$) for statistical accuracy comparisons and the option-data subsample $\mathcal{T}_{\mathrm{opt}} \subset \mathcal{T}_{\mathrm{oos}}$ for straddle return comparisons (Section 3).
  - We denote by $\bar{d}$ the sample mean of $d_t$ over $\mathcal{O}$ and compute two-sided $p$-values under the standard normal distribution as $p = 2\{1 - \Phi(|\mathrm{DM}|)\}$.
  - Under this sign convention, $\mathrm{DM} < 0$ indicates that forecast $A$ achieves a lower average loss than forecast $B$. Throughout the paper, "statistically significant" refers to significance at the 5% level ($p < 0.05$).
- **Newey–West long-run variance.** Following Newey and West (1987), we estimate the long-run variance $\hat{\Omega}$ using Bartlett kernel weights with lag truncation parameter $q_{\mathrm{NW}}$:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\mathrm{NW}}}\Big(1 - \frac{j}{q_{\mathrm{NW}}+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{|\mathcal{O}|}\sum_{t=j+1}^{|\mathcal{O}|}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

**E13.** Decision: regenerate the remaining DM statistics.
Problem: `dm_test` now divides $\hat{\gamma}_j$ by $|\mathcal{O}|$, as displayed. The accuracy tables and macros were regenerated on 2026-10-18, but the DM statistics and significance stars in the appendix tuning and study tables are read from stored study outputs computed with the old divisor $|\mathcal{O}| - j$, so they predate the change and must be regenerated.

  - For the Diebold–Mariano test, we set the bandwidth according to the plug-in rule $q_{\mathrm{NW}} = \lfloor 4(|\mathcal{O}|/100)^{2/9} \rfloor$ (its numerical value on $\mathcal{T}_{\mathrm{oos}}$ is reported in Section 6.1).
- **HAC $t$-statistic for mean daily returns.** For any daily return series $r_t$—either the return of an individual trading strategy or the return differential $\Delta_t$ between two strategies—we compute the unadjusted Newey–West $t$-statistic

$$t = \frac{\bar{r}}{\sqrt{\hat{\Omega}_r/|\mathcal{O}|}},$$

  where $\hat{\Omega}_r$ is the Newey–West variance estimator above evaluated on $r_t$ in place of $d_t$, using lag truncation $q_{\mathrm{NW}} = \lfloor 1.5\,|\mathcal{O}|^{1/3} \rfloor$ (its numerical value on $\mathcal{T}_{\mathrm{opt}}$ is given in Section 7.2).
- **Multiple-comparison procedures.** For joint comparisons across predictor sets and model families, one may also consider the model confidence set of Hansen et al. (2011), the superior predictive ability test of Hansen (2005), and the StepM procedure of Romano and Wolf (2005).

**E14.** Decision: drop the sentence, or report the set.
Problem: the model confidence set, the superior predictive ability test, and StepM are named, and the paper reports none of their outputs.
This bullet names three procedures and says they are for the comparison of predictor sets: the model confidence set, which returns the set of models not beaten at a stated confidence level; the superior predictive ability test, which returns a $p$-value against a benchmark; and StepM, which steps through the models and rejects some of them. No table in the paper states an MCS membership, an SPA $p$-value, or a StepM rejection. The next bullet says the main tables are point estimates and contain no bootstrap intervals, which is the kind of output those three procedures produce. The comparisons that are printed are Diebold–Mariano statistics on QLIKE and on squared error.

- **Reporting convention.** All main tables report point estimates accompanied by pairwise Diebold–Mariano or HAC $t$-statistics, without bootstrap confidence intervals.

# J Notes to the authors (not part of the paper)

- **Accuracy against the benchmark:** 0 of the 12 pooled intraday forecasts have lower QLIKE than the daily HAR benchmark, 0 significantly (Diebold–Mariano, 5% level); closing-interval tests are left out of this count (Appendix G).
- **Data issue to fix before final numbers:**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on all 122 trade days after November 2023;

    **S4c.** Decision: rerun needed under the causal rule (note S4a).
    Problem: this item describes the old behaviour of the printed forecasts of 6–7 October 2026 (full-sample mode, flags read after the fill). Commit b29774a (15 October 2026) sets CAUSAL_MODE_INFLATION and ACTIVE_FROM_OBSERVED in src/backtest/executor.py, so after the rerun the flags come from observed prints. The code comment counts 123 FOMC days, not 122.
  - VIX, volume-demand and FOMC feeds end early and are carried forward without limit into 2024.
- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models.
- **Pending:** the pooled random forest (full table).
- **E15, sample and window still open** (2026-10-16). Hyperparameters: one configuration per model, chosen once on the pre-sample and frozen on the evaluation span. The daily HAR OLS stays outside the scheme. Not applied to Table 1.
  - Open choice: all 48 intervals, overnight included, or the 13 regular-hours bars (end stamps 10:00–16:00). Window: 24,000 intervals or 96,000 intervals (2,000 trading days on the 48-bar grid).
  - Calendar (note E9): columns after the overnight indicator are out of the specification, including the products of the target HAR components with the opening- and closing-interval indicators. The overnight indicator stays only if the overnight intervals stay in the training sample. On the 13 regular-hours bars it equals 1 only on the 16:00 bar. Not applied to the printed column counts or Table 1.
  - Shorter window, 16:00 rows only (closing-interval window study, 2026-10-03; trees refit every 10 sessions and the linear penalties are re-chosen, so this is not the frozen rule). LightGBM QLIKE 0.1110 at 24,000/48 sessions and 0.1006 at 2,000 sessions (DM 2.71). Ridge 0.1194 and 0.1027 (DM 2.94). Lasso 0.0998 and 0.1006 (DM −0.36).
  - A 48-bar pool against a one-bar fit, HAR plus calendar, 2012–2024. The pool preferred the shorter window; the one-bar fits preferred 2,000 days. At the 16:00 bar the one-

bar QLIKE gain was 6–8% on the shorter window and 12–13% at 2,000 days. The full predictor set on one bar was worse on the shorter window (+4.1%, 8 bars worse) and flat at 2,000 days (+0.5%).

  - The 13 regular-hours bars as one model against the 48-bar pool, 2,000 days, mean over buckets. QLIKE lower by 1.75% / 1.86% / 1.85% (ridge / elastic net / lasso); at the close, 5.5–6.0%. The 15:30 sign trade, crossed, matched the pool (0.70 / 0.62 / 0.68 against the pool's 0.72 / 0.68 / 0.73). It moved only for a fit on the 16:00 bar alone (QLIKE 8.6–9.5% below the pool; crossed Sharpe 1.00 / 0.92 / 0.85, lasso / elastic net / ridge).

- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.